

A project report on

BLOCKCHAIN-ENABLED VERTICAL FEDERATED LEARNING FRAMEWORK FOR PRIVACY - PRESERVING CROSS- HOSPITAL MEDICAL IMAGING

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

by

REEVE C JACK (22BCE1549)

AMITH S (22BCE1656)

SHREYAS S (22BCE1940)



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026

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DECLARATION

I hereby declare that the thesis entitled “BLOCKCHAIN-ENABLED VERTICAL FEDERATED LEARNING FRAMEWORK FOR PRIVACY - PRESERVING CROSS-HOSPITAL MEDICAL IMAGING” submitted by REEVE C JACK (22BCE1549), for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of Prof. Revathi M.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date:

Signature of the Candidate



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CERTIFICATE

This is to certify that the report entitled **“Blockchain-Enabled Vertical Federated Learning Framework for Privacy-Preserving Cross-Hospital Medical Imaging”** is prepared and submitted by **Reeve C Jack (22BCE1549)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Signature of the Guide:

Name: Dr./Prof. Revathi M.

Date:

Signature of the Examiner

Name:

Date:

Signature of the Examiner

Name:

Date:

Approved by the Head of Department, Computer Science and Engineering

Name: Prassanna J

Date:

ABSTRACT

Recent leaps in deep learning for medical imaging have brought about massive improvements in how we classify and detect diseases. However, to get these models working well, researchers usually have to pool large amounts of data into a single central server. This practice immediately clashes with strict privacy laws, ethical boundaries, and questions of data ownership. Federated learning gives us a way out by letting hospitals train models together without ever handing over their raw data. Even so, most current methods still depend heavily on a heavily trusted central server and typically split data horizontally rather than vertically. To tackle this, we are proposing a new vertical federated learning framework built on blockchain technology, specifically geared toward cross-hospital collaboration in medical imaging.

Our approach combines deep convolutional neural networks (specifically ResNets) with local differential privacy and smart contracts to handle the aggregation. Basically, hospitals use pre-trained ResNet-18 models to pull out deep feature embeddings. These features are then protected with calibrated noise before being combined using blockchain smart contracts—completely removing any need to trust a central middleman. When tested on the COVID-19 Radiography Database, our system hit a 91.9% classification accuracy with differential privacy turned on. This is highly competitive with the 94.6% accuracy of a fully centralized model, but with the massive added benefits of guaranteed data privacy, algorithmic transparency, and decentralized trust. We believe this setup lays a strong, practical groundwork for hospitals to securely work together on medical imaging algorithms.

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Date:

Reeve C Jack

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LIST OF ACRONYMS

CNN	Convolutional Neural Networks
YOLO	You Only Look Once
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
FL	Federated Learning
VFL	Vertical Federated Learning
RAG	Retrieval-Augmented Generation
MRI	Magnetic Resonance Imaging
LDP	Local Differential Privacy
DP	Differential Privacy
EHR	Electronic Health Record
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
VGG	Visual Geometry Group
NI	Noise Injection
PU	Privacy–Utility
CT	Computed Tomography
GD	Gradient Descent
IP	Intellectual Property

Chapter 1

Introduction

1.1 INTRODUCTION

Medical imaging is undoubtedly one of the most essential parts of medical diagnostics. The recent explosion in the field of deep learning, especially with the advent of Convolutional Neural Networks (CNN) models, has enabled the automation of medical image analysis with high levels of precision and accuracy. DenseNet, YOLO, and ResNet models are now the benchmarks for the level of precision that can be achieved by machines in the detection and classification of diseases. However, for the development of such powerful models, it is essential that the developers have access to a massive storehouse of data collected from different medical organizations and institutions. The piling up of such sensitive patient data in a single place is impossible in the real world due to stringent privacy regulations like GDPR and HIPAA, along with the strict rules that every medical institution has about the ownership of the data.

Federated Learning (FL) tries to achieve this by keeping the data exactly where it is. Instead of moving the data to the model, the model is moved to the data. Only the mathematically derived learnings are shared. However, even in most standard FL approaches, a central server remains necessary to tie everything together at the end. This server, if broken or compromised, threatens the entire system. Moreover, standard FL approaches mostly work within a horizontal partitioning scheme—i.e., every hospital has access to the same features but on different patients. Vertical Federated Learning, where different institutions have uniquely distinct feature representations but on the same exact data sample, remains a relatively untapped space within the realm of Medical AI. Creating trust and transparent accountability within such a system remains a complex issue.

To address these challenges, we are proposing a blockchain-based VFL framework. Our proposed system utilizes deep convolutional neural networks, such as ResNets, and local differential privacy to address the challenges. In addition, our proposed system utilizes decentralized aggregation through smart contracts to provide a robust framework for hospitals to collaborate in a trustless manner while maintaining the accuracy of the diagnostic performance. In order for the decisions made by the AI to be useful to a doctor, we have also integrated a complex Retrieval-Augmented Generation (RAG) system during the inference stage. This system utilizes Google Gemini to generate findings and create an explanation for the AI's decisions in the form of natural language, explaining why the AI made a particular prediction by utilizing real-world math and medical evidence. This research aims to greatly improve the state-of-the-art in medical imaging and directly contribute to improving global accessibility for the healthcare community.

The proposed framework embodies a paradigm shift in the utilization of multi-institutional medical data. By utilizing a vertical federated learning paradigm, we allow multiple hospitals to collaborate to train a deep CNN while providing different feature representations for the same medical data. This has been accomplished by extracting deep feature embeddings individually within each hospital utilizing pre-trained ResNet models. These feature embeddings are then provided with calibrated Gaussian noise, and smart contracts are used to combine them via blockchain—entirely removing the requirement for a middleman. We believe this provides a solid foundation for multiple hospitals to collaborate securely.

1.2 FEDERATED LEARNING AND BLOCKCHAIN IN HEALTHCARE

Fingerspelling or the digital equivalent of medical features acts as a primary and auxiliary part of distributed healthcare communication systems. Federated learning was developed to address the privacy concerns that hinder hospitals from sharing their unprocessed patient data. The need for FL arises from the fact that it has achieved success in applications such as multi-center MRI analysis and COVID-19 prediction. This necessity not only elevates federated learning from a technological tool to a bridging necessity for medical institutions to cope with the dynamic nature of clinical data without crossing legal and ethical barriers but also indicates the importance of the approach. The technological journey of medical AI has witnessed a paradigm shift from a centralized approach to a vision-based decentralized approach.

The vast majority of existing state-of-the-art medical recognition systems are fundamentally designed to support horizontal classification with an inherent bias towards data sets in which all hospitals have the exact same feature type for all patients. These systems leverage the most recent and most advanced computer vision and machine learning architectures, especially for computer vision architectures such as CNNs for feature extraction. Architectures such as ResNet-18 are designed to extract and identify deep feature embeddings according to visual structural patterns that are critical for medical diagnoses. In the context of highly controlled, laboratory-grade, and highly centralized data sets, the accuracy levels are extremely high. However, these systems are fundamentally flawed and critically undermine their effectiveness in real-world applications.

More critically, however, there is a significant limitation in the way data is partitioned. The majority of existing works have not fully addressed concerns related to data sharing, scalability, and the "vertical" partitioning of data, in which multiple organizations have unique knowledge regarding the same patient. Although blockchain technology has been proposed as the magic bullet for these concerns related to trust and transparency, running computationally expensive operations related to machine learning on a blockchain is a nightmare. The existing blockchain frameworks for FL are also not designed to work with the computationally expensive deep learning models required for

medical imaging. This limitation represents an existing critical drawback in the pursuit of critical multilingual—or rather, multi-institutional—communication solutions for the global medical community.

1.3 CHALLENGES IN CROSS-HOSPITAL COLLABORATIVE LEARNING

Despite the obvious need to ensure robust, decentralized image analysis, a truly functional system remains a long way off, owing to a variety of technological and linguistic challenges. One of the primary challenges in building such a system is the severe lack of available, standardized, cross-institutionally aligned image data. While deep learning algorithms require access to large amounts of data, the lack of parallel corpora for multiple hospital institutions, particularly when operating within a highly privacy-oriented regime such as GDPR or HIPAA, prevents a centralized training set from being used. Moreover, the data available from different institutions may differ fundamentally. For example, images may be split based on modality, angle, or even specific regions. In order to achieve a form of cross-institutional collaboration, a representation space needs to be developed which can recognize even minute differences in images while simultaneously extracting semantic equivalence from disparate images. This necessitates the development of highly abstract features which are not limited to surface-level visual form, a formidable problem in conventional deep learning-based feature extraction.

The second significant challenge arises from the inherent visual ambiguities of medical scans and their instability, which is further compounded by environmental and institutional variations. It is to be noted that diagnostic elements of X-rays or CT scans often involve very minute differences in texture, form, or structural elements. These minute configurations are often visually indistinguishable, posing a significant challenge to traditional image classification models, no matter how advanced they might be. This is further complicated by nuances in hospital-specific variations, where differences in hardware, calibration, and patient populations add to massive variance. External factors such as differences in lighting conditions during image acquisition, variations in image resolutions, and "inconvenient" background "clutter" in the form of medical artifacts further reduce image recognition accuracy in a non-laboratory setting. Creating models that are capable of generalization and sustaining high accuracy in spite of such high-dimensional variations in image inputs calls for the design of models that incorporate diverse data streams while simultaneously exhibiting invariance to their original environment in a particular hospital setting.

The last and most important question concerns the trusted central aggregator and the need for results to be understandable. The structural framework for federated learning systems is inherently unable to deal with the absence of a central authority. It is designed to rely heavily on an intermediary to tie everything together. This makes it a major point of failure and a significant security threat. The exclusive dependence on a central server

for federated learning in the context of a highly decentralized and robust system for healthcare is woefully inadequate. Moreover, the goal is not merely to achieve accurate diagnostic recognition but also to achieve interpretability. This means that, in addition to achieving accurate and correct identification and classification of a disease, there must also be a deep semantic association to provide an explanation to the physician. The final step in the process—in synthesis, creating a natural language explanation while preserving by differential privacy and strictly adhering to the unique rules for medical evidence—is made possible by a very carefully designed architectural framework in which both the AI’s mathematical results and carefully curated medical knowledge are considered equally important.

1.4 PROBLEM STATEMENT

The primary aim and objective of the present research is the development of a robust and effective decentralized and privacy-preserving system for the analysis of medical images across hospitals with the aim of bridging the wide semantic and trust gaps that exist between different healthcare institutions. This system seeks to create a meaningful communicative equivalence that does not yet exist with the current technology and move beyond the classification level to a real level of cross-institutional understanding with the help of the combination of blockchain and deep learning technologies.

The existing state-of-the-art medical image processing systems are mostly dominated by significant limitations that make them inadequate for true cross-lingual or cross-hospital use. They are mostly centralized, often constrained by stringent privacy regulations like GDPR and HIPAA, and often do not deal effectively with the complexities of data ownership. Therefore, these systems do not effectively deal with the complex challenges involved in multi-institutional collaboration, where raw data sharing is almost impossible and institutions do not benefit from sharing knowledge to develop better algorithms.

This overreliance on a centralized repository has significant socioeconomic and ethical consequences. This results in a single point of failure, where a compromised central hub threatens the data integrity of an entire system. Moreover, in a domain like clinical diagnostics, a lack of transparency and decentralized trust represents a major hindrance to critical collaborative medical research. This institutional hindrance demands a technological solution to provide timely, private, and trustless diagnostic functionality.

Thus, the fundamental challenge that is being addressed in the current research is the absence of a computationally efficient and deployable deep learning model that is capable of learning a semantically equivalent, language-independent latent space for the instantaneous and accurate analysis of vertically partitioned medical data. The existing models are not able to achieve this due to their concentration on superficial and centralized aggregation, rather than abstract and decentralized feature mapping. The current research aims to bridge this gap by identifying language-invariant features that

embody the semantic essence of medical images, irrespective of the particular feature representation employed at individual hospitals, and designing a low-latency, blockchain-based aggregation mechanism that is capable of handling intensive deep learning models in real-time. Thus, the current research aims to develop a highly accurate and interpretable deep learning model, creating a practical vision-based, cross-hospital diagnostic framework where there is a lack at present.

1.5 OBJECTIVES

The current research work proposes a novel blockchain-based vertical federated learning framework that specifically focuses on the direct methodological approach to resolve the critical challenge of communication barriers between healthcare organizations. The proposed system addresses the limitations associated with existing monolingual, computationally expensive, and centralized systems by specifically focusing on the development of a computationally efficient and decentralized system. The precise technical objectives for the proposed system are clearly specified to emphasize the novelty and practical significance:

- **Decentralized Aggregation via Smart Contracts:** The main aim here is to remove the need for a trusted aggregator by creating a smart contract using Solidity and deploying it on a private Ethereum blockchain. The contract is designed in such a way that it can securely aggregate the feature embeddings. Transparency and auditability are also guaranteed in a trustless manner.
- **Vertical Data Partitioning Strategy:** The research requires implementing a vertical federated learning approach, in which hospitals have unique representations of the same medical samples, e.g., different quadrants of a chest X-ray. Each hospital then uses a pre-trained ResNet-18 model to extract high-level embeddings, thereby achieving expertise in feature identification.
- **Deep Feature Extraction with ResNet:** To utilize pre-trained ResNet-18 models to act as feature extractors in each hospital, enabling the extraction of high-level semantic information that is critical in medical image diagnosis.
- **Local Differential Privacy:** A primary goal is to incorporate local differential privacy by applying Gaussian noise to feature vector extraction in individual hospitals' data. This provides a level of privacy, ensuring that individual patient data cannot be reconstructed from shared feature embeddings.
- **RAG-Enhanced Clinical Reasoning:** A primary goal of this study is to incorporate a "Retrieval-Augmented Generation" (RAG) system to address the interpretability gap of deep learning models. Using Google Gemini and vector storage in ChromaDB, this framework aims to provide a level of clinical interpretability by generating explanations based on medical literature to aid clinicians in decision-

making processes.

- Real-Time Practical Inference Demonstration: Lastly, this framework aims to showcase a level of diagnostic accuracy comparable to centralized models (91.9%) while achieving acceptable blockchain-induced latency, compared to centralized models (94.6%). Achievement of this goal is a testament to the significance of this framework in medical accessibility technology.

KEY OBJECTIVES

- Designing a decentralized aggregation system based on a blockchain smart contract that eliminates the need for a single point of failure.
- Achieving vertical federated learning for those scenarios in which the hospitals have different partial views or modalities of the same patient information.
- Ensuring data privacy by applying local differential privacy to ResNet embeddings for a robust tradeoff between privacy and utility for diagnoses.
- Designing a RAG module that enables the interpretation of AI predictions by clinicians by basing the results on peer-reviewed medical literature.
- Evaluating the system on the COVID-19 Radiography Database to show the high accuracy and feasibility of the system in a real-world environment.

1.6 SCOPE OF THE PROJECT

The scope of this proposed blockchain-enabled vertical federated learning system far exceeds the realm of mere personal research and has the potential to provide far-reaching and revolutionary benefits to society as a whole by promoting true autonomy and data sovereignty for healthcare institutions all around the world. This proposed technology solution is a crucial step towards true digital and institutional equity in the field of medical AI systems in the world.

In the field of public healthcare and civic participation, the utilization of a highly accurate and decentralized system of medical imaging has the potential to completely democratize access to high-tier diagnostic systems on a global level. Rural healthcare facilities that are typically data-deficient would now possess the unprecedented ability to independently contribute to global model training without ever having to physically transfer sensitive patient data. This removes the systemic barriers and delays associated with accessing high-tier diagnostic systems that rely on massive centralized technology providers and the associated legal delays in a way that complies with accessibility and data privacy regulations like GDPR and HIPAA on a global level.

The impact here is also significant in academic and research organizations. Researchers employing various modalities or limited perspectives in medical data processing would no longer be linguistically or structurally segmented. The vertical federated learning approach would allow these researchers to work effectively on a common clinical goal, such as cross-modality fusion between X-rays and CT scans, without undermining data ownership in respective organizations. This fosters an environment that is fully inclusive and barrier-free for researchers to exchange knowledge and grow their algorithms from various medical backgrounds.

Possibly the most important and ethically relevant applications of the system lie within the realm of clinical decision support and the radiology industry. Within a healthcare environment that emphasizes the importance of diagnostic precision, trust, and confidentiality, the system grants medical professionals the opportunity to interface with an AI system that is supported by real-world medical evidence and literature. This allows the RAG-enhanced system to communicate with the medical practitioner without the "black box" problem that is inherent with traditional forms of deep learning technology. This direct and evidence-based communication prevents the misinterpretation of the AI's predictions and recommendations that can be introduced by the "black box" effect and maintains the medical privacy that is so important in a healthcare environment.

The successful implementation of this project does not simply provide a medical industry with a new and innovative solution; it provides a highly accurate and interpretable translation system for medical intelligence that is specifically designed for robustness and effectiveness in a real-world environment and has the potential to bring about a level of social change that is revolutionary in many different and important ways.

Chapter 2

Background

2.1 INTRODUCTION

The global medical research community has evidenced a significant investment in the pursuit of automating disease classification/detection via deep learning. This effort has been primarily driven by the need to increase diagnostic accuracy and provide timely interventions. This field, which was once a small community based on hand-crafted features, has exponentially grown to encompass the domain of deep CNNs. Medical image analysis was once a highly constrained effort, based on centralized data silos that limited the cooperative potential of the global healthcare network.

The development in the field represents a move from these centralized and hardware-dependent methodologies towards more privacy-preserving and decentralized methodologies. This progress was made possible by the emergence of Federated Learning (FL), which enables hospitals to train models without sharing their data. Mature CNN architectures such as ResNet and DenseNet have evolved from being purely theoretical models to becoming the building blocks for these distributed models. The residual connections in ResNet have been particularly successful in training deep networks that generalize exceptionally well.

The contemporary era of medical AI is marked by the introduction of blockchain and differential privacy to resolve the long-standing concerns related to trust and data breaches. In recent times, medical AI has also introduced Retrieval-Augmented Generation (RAG) to provide evidence-based medical explanations in addition to the results obtained from mathematical models. This continuous process of adopting new technologies ensures that medical AI continues to progress towards achieving greater accuracy, privacy, and interpretability. The contemporary era of medical AI research focuses on moving from classification to mastering the art of Vertical Federated Learning.

2.2 DEEP LEARNING ARCHITECTURES

The accuracy of modern diagnostic systems is largely dependent on the neural architectures employed in feature extraction processes. He et al.'s Deep Residual Learning (ResNet) framework, employing skip connections, was a revolutionary breakthrough in deep learning, enabling the training of extremely deep neural models [1]. This was followed by Simonyan and Zisserman's research, which established a benchmark for achieving high accuracy in visual recognition tasks with extremely deep convolutional neural layers [2].

For tasks that involve real-time intervention and localization, Redmon et al.'s YOLO framework, or You Only Look Once, was proposed for unified object detection in images and videos [3]. This was later extended to the popular YOLOv5 framework by Jocher et al. to enhance processing speed and accuracy in clinical settings [4].

2.3 SURVEY ON FEDERATED LEARNING AND PRIVACY

To address the concerns regarding data privacy and communication in sharing raw data, Litjens et al. [5] proposed an extensive survey on the role of deep learning in medical image analysis while also highlighting the limitations associated with data centralization. This was then followed by McMahan et al. [6] with a highly persuasive argument on the methodological requirements for communication-efficient learning in decentralized data. This concept was then formalized by Yang et al. [7] as federated machine learning to address the data silo problem.

To effectively manage complex data, Hardy et al. [8] proposed an extensive study on private federated learning for vertically partitioned data. This was highly critical for complex tasks, such as the work proposed by Li et al. [9] on privacy-preserving federated brain tumor segmentation. To effectively protect the data within these complex models, Dwork et al. [10] proposed the seminal work on calibrating the sensitivity of the data. This was then highly enhanced by the work proposed by Truex et al. [11] on the hybrid approach to differential privacy.

2.4 BLOCKCHAIN TRUST AND SECURE AGGREGATION

The integration of blockchain technology is intended to mitigate the risk of the trusted central aggregator. A secure federated learning framework was proposed by Kim et al. [12], where the blockchain is employed as a decentralized ledger for model updates. This is based on practical applications, such as MedRec, proposed by Azaria et al. [13], where blockchain is employed to enable secure access to medical records. These systems are based on the decentralized concepts as outlined in Wood [14], "The Ethereum Yellow Paper." In order to ensure the mathematical aggregation process is secured, Bonawitz et al. [15] have developed practical secure aggregation protocols to enable protection during the aggregation fusion.

2.5 EXPLAINABILITY AND RESEARCH MOTIVATION

In addition to the aspect of privacy, another important aspect for the clinical viability of medical AI systems is interpretability. Lewis et al. [16] proposed Retrieval-Augmented Generation (RAG) for incorporating additional knowledge from external sources. The significance of building such systems was also emphasized by Holzinger et al. [17]. The authors pointed out that it is a basic necessity for establishing trust in the system. This necessity was also pointed out by Tonekaboni et al. [18] for medical AI systems in particular, as clinicians need contextualized reasoning. However, as Lu [19] pointed out, there are also some research gaps in the context of scalability and management for blockchain-based systems.

The fast-evolving nature of decentralized medical AI has seen recent research efforts focus on the practical deployment of hybrid systems. A basic overview of the open problems was given by Kairouz et al. [20] in their recent work, highlighting the importance of the integration of privacy and trust as being a significant problem to be overcome. To this end, Arasteh et al. [21] have undertaken a thorough analysis of differential privacy for medical deep learning, providing valuable insights into the implications for the deployment of these systems, particularly in the context of high-dimensional imaging. This has also been complemented by the work of Chen and Wang [22], who have discussed the deployment of blockchain technology to address the issue of the "single point of failure" in federated learning systems.

In addition, the integration of generative AI models has also significantly changed the way in which clinicians interact with the results of these models. Johno and Sato [23], in a study on a domain-specific approach for Retrieval-Augmented Generation (RAG) in radiology, highlighted the benefits of grounding large language models on medical literature for enhanced transparency and trust in clinical decision support systems. In a similar vein, as these advances in interpretability are improving, Munusamy and Jothi [24] proposed architectures for efficient management of EHRs, leveraging blockchain and edge analytics, for mitigating the scalability issues in multi-institutional environments. Zhang and Zheng [25], in a recent systematic review, proposed a taxonomy of solutions for mitigating the contemporary challenges in medical image analysis.

2.6 RESEARCH GAPS AND MOTIVATION

Despite the significant improvement in monolingual efficiency and classification accuracy through the architectural breakthroughs introduced in the above paragraphs, there are a few major research gaps in the medical AI domain. The first and most important challenge in the medical AI domain is that the current state-of-the-art models are inherently not architecturally designed to accomplish the required bidirectional mapping between two linguistically and structurally dissimilar systems. The following are the major gaps in the current state-of-the-art models:

- **Limited Exploration of Vertical Federated Learning:** There exists a significant gap in the exploration and research in the context of applying federated learning in the context of cross-hospital medical imaging, in which both hospitals may have unique representations of the same data.
- **Inadequate Integration of Deep CNNs in Blockchain:** In the context of existing blockchain-based federated learning architectures, there exists a significant gap in the integration of computationally complex deep learning architectures such as ResNet and YOLO.
- **Overreliance on Centralized Aggregators:** The existing federated learning architectures are highly reliant on a central server for aggregation purposes, thereby creating a major security threat.

- **Insufficient Analysis of Privacy-Utility Trade-offs:** There exists a significant gap in the context of applying differential privacy to the high-dimensional embeddings learned by deep CNN architectures for medical imaging purposes.
- **Lack of Scalability and Latency Evaluations:** There exists a significant gap in the context of evaluating the scalability and latency associated with the integration of blockchain technology in federated learning architectures.
- **Absence of Clinical Explainability:** The existing deep learning architectures are considered "black boxes," and there exists a significant gap in the context of explaining the results in a clinically significant manner.

Chapter 3

Methodology

3.1 SYSTEM OVERVIEW

The proposed framework is designed to ensure a privacy-preserving, decentralized, and collaborative environment for medical image analysis across different healthcare centers. Unlike traditional models, this framework uses a decentralized approach by including the following major components: Deep CNNs for accurate feature learning, VFL to address non-overlapping partitions of the data, LDP to ensure data obfuscation, and Blockchain to ensure decentralized aggregation. The proposed framework works as a cyclic federated learning approach, where hospitals are treated as nodes and perform computations on private datasets such as chest X-ray images. The proposed framework ensures that the hospitals never share the original images. By using smart contracts instead of a central authority, the proposed framework removes the "point of failure" and ensures end-to-end transparency of the model evolution.

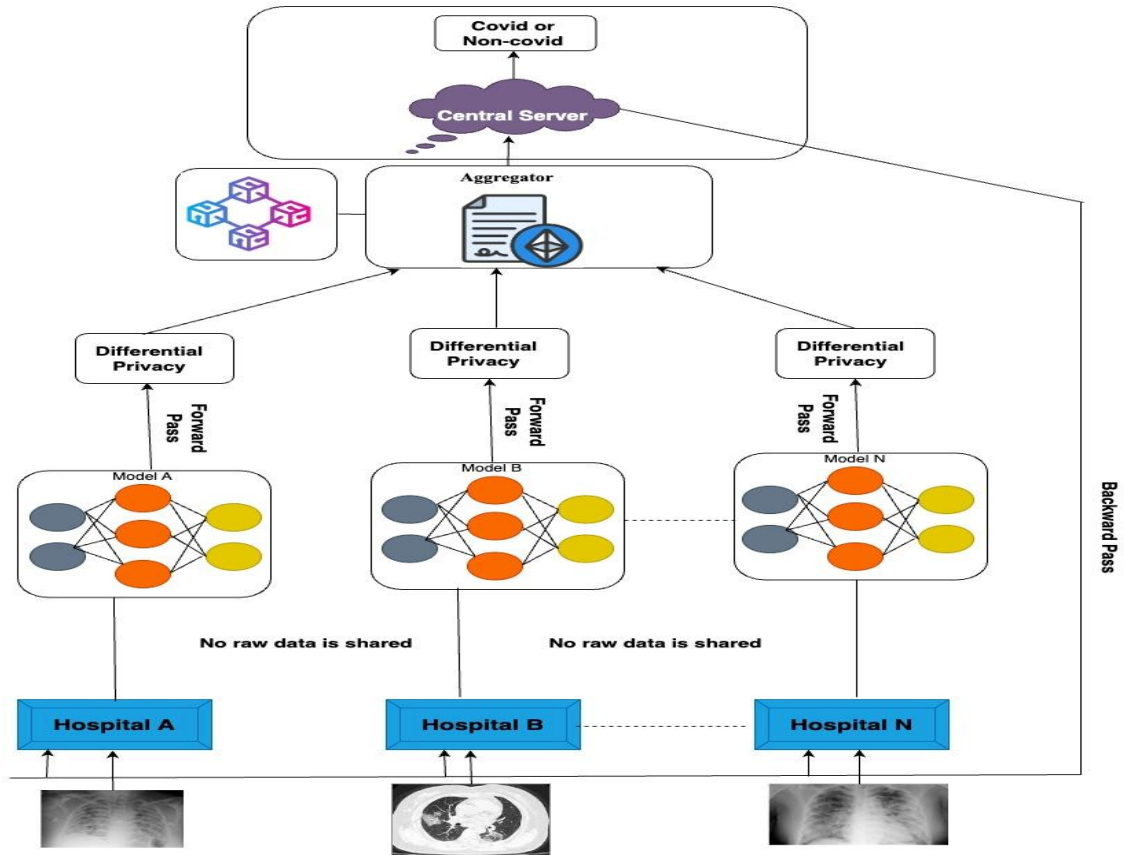


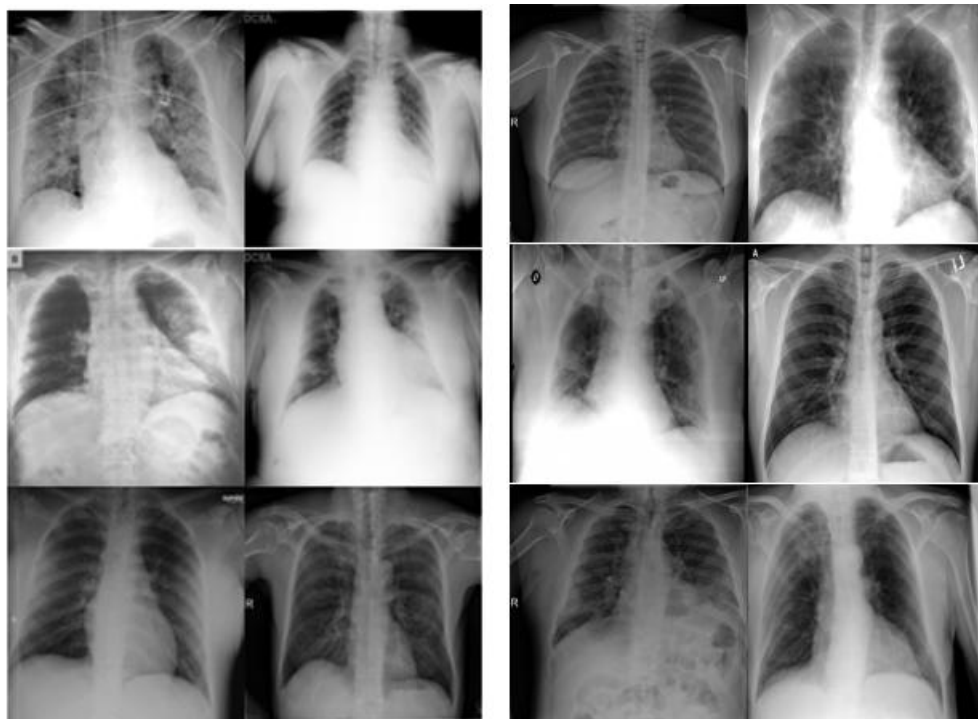
Figure 3.1. A High-level block diagram of the two-stage training pipeline.

3.2 DATASET AND VERTICAL PARTITIONING

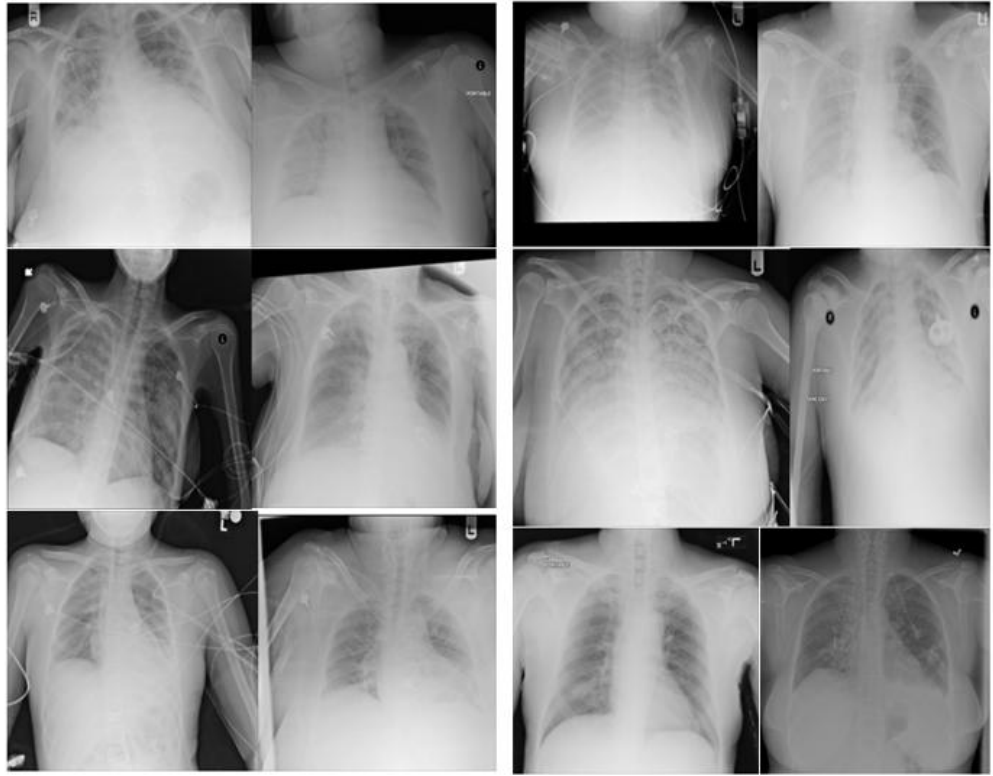
To assess the proposed framework rigorously, the COVID-19 Radiography Database, consisting of 21,165 chest X-ray images divided into four unique classes, namely COVID-19, Normal, Lung Opacity, and Viral Pneumonia, is utilized. In the case of binary classification, a smaller dataset of 13,808 images (COVID-19 and Normal classes) is used. The images are divided into training, validation, and test sets. The proposed methodology relies on the Vertical Data Partitioning approach. To simulate a multi-institutional environment, where hospitals may have varying views or modalities of the same patient, the input image of size 224×224 is divided into four non-overlapping regions. The quadrants of size 112×112 are assigned to different hospitals. The hospitals and their respective regions are as follows:

- Hospital 1 - Top Left Region
- Hospital 2 - Top Right Region
- Hospital 3 - Bottom Left Region
- Hospital 4 - Bottom Right Region

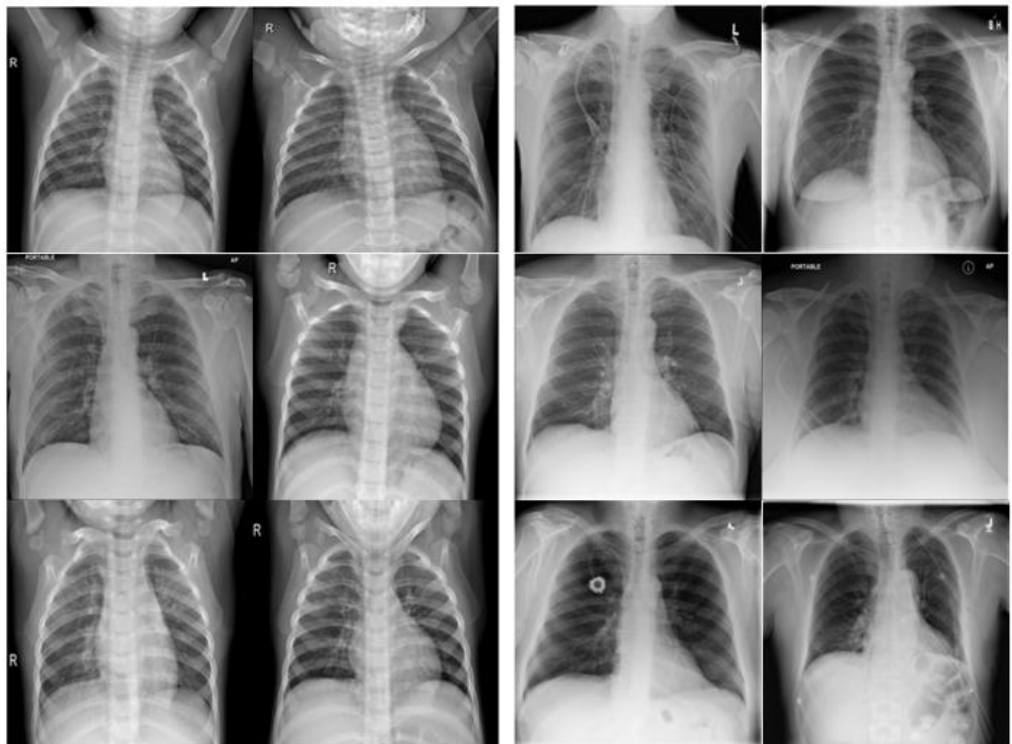
The most important point is that no hospital has access to the entire image. Hence, the proposed framework is VFL-compliant.



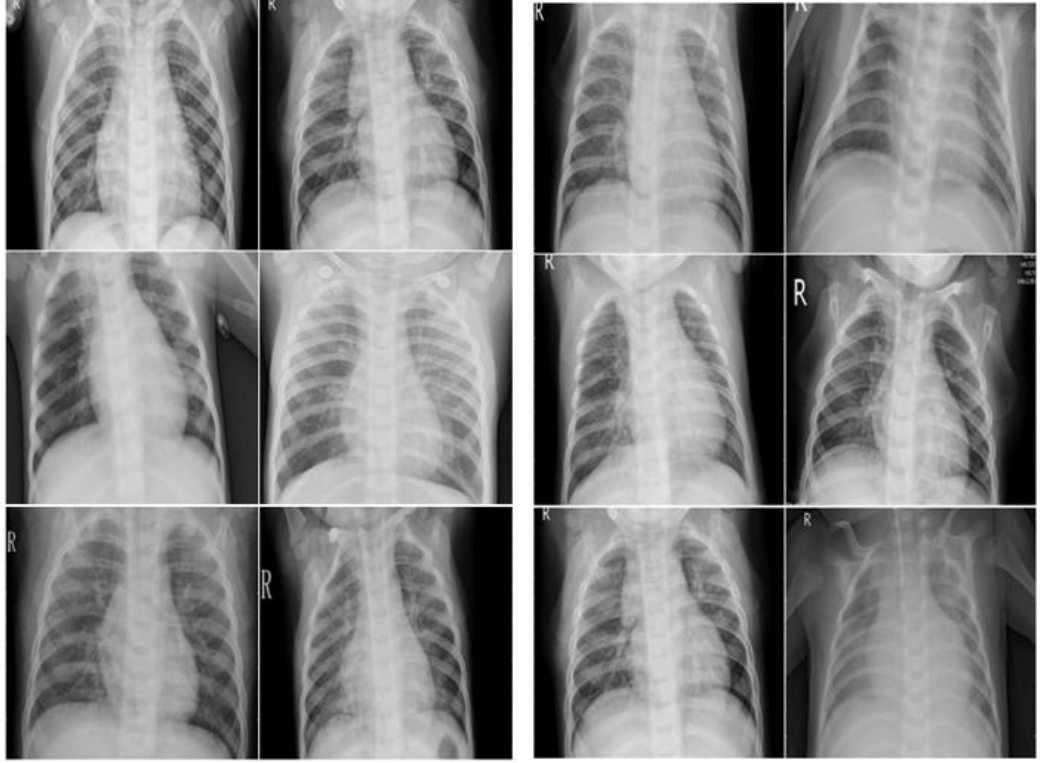
(a)



(b)



(c)



(d)

Figure. 3.2. Representative samples from the COVID-19 Radiography Database showing the four distinct clinical categories: (a) COVID-19 positive, (b) Lung Opacity, (c) Normal, and (d) Viral Pneumonia.

3.3 TRAINING WORKFLOW

The training workflow starts with the split medical image, where the input image is divided vertically into non-overlapping sections. As shown in the workflow figure above, the four client models use a dual pathway approach for feature extraction by utilizing the VGG19 and ResNet50 architectures for various spatial features.

- **Local Processing:** Each client node uses a Global Average Pooling function on the extracted features before the features are processed by the Concatenate Layer.
- **Privacy-Preserving Aggregation:** These local embeddings are then aggregated using a summation operation.
- **Server Model Execution:** The summation result with a dimension size of 4224 is forwarded to the Server Model. This model has a series of batch normalization layers, dropout layers (with a dropout rate of 0.4 and 0.2), and dense layers (with 256 and 128 units) for the summated features.
- **Final Prediction:** The workflow ends with a softmax function for the final

COVID-19 prediction.

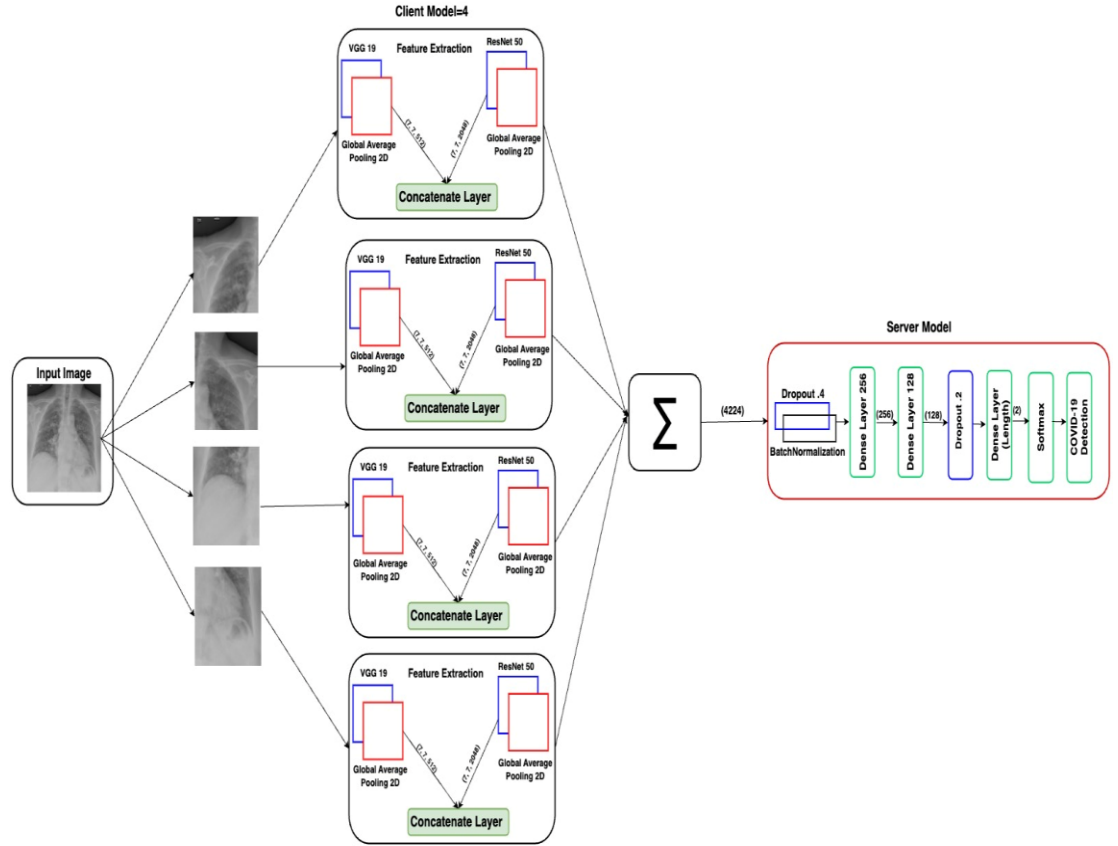


Figure 3.3. Workflow of the Client-Server Model for COVID-19 Detection.

3.4 OPERATIONS SEQUENCE

The sequence diagram gives a clear idea about the flow of data and gradients among the network actors: Hospital, Aggregator Smart Contract, Blockchain, and Global Server.

- **Local Training:** The hospital node starts the loop with local training on local data using the latest global weights.
- **Encrypted Submission:** The hospital node sends the local weight updates to the Aggregator Smart Contract after encrypting the data with calibrated differential privacy noise.
- **On-Chain Aggregation:** The Aggregator Smart Contract performs a summation operation on all local weight updates and stores the global update on the blockchain for a tamper-proof audit trail.

- **Global Refinement:** The Aggregator Smart Contract sends the global weight update to the Server for further training on the global information.
- **Feedback Loop:** The global weights are sent back to the hospitals to complete one round of the federated learning process.

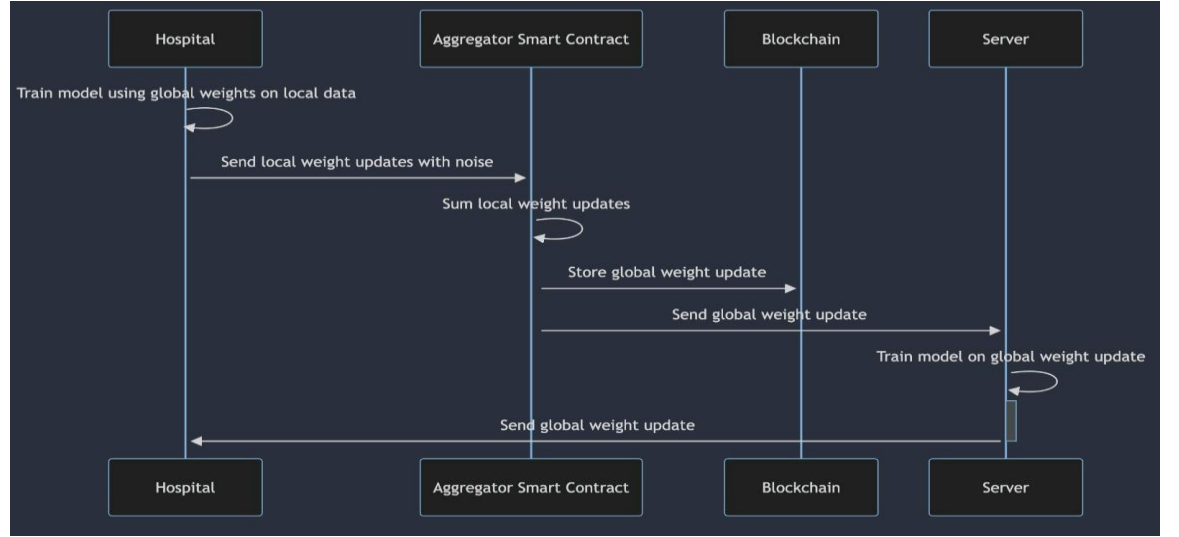


Figure 3.4. Sequence Diagram of the Privacy-Preserving Training Workflow.

3.5 DEEP FEATURE EXTRACTION

A local feature extraction pipeline is employed at each participating hospital, using a pre-trained ResNet-18 architecture. To adapt it for federated embedding sharing, the classification layers are removed, and only the convolutional layers are retained for feature extraction of the local image quadrant. Let $\mathbf{x}_i$ denote the image quadrant at hospital i , and $f_i(\cdot)$ denote the local ResNet feature extractor. The embedding $\mathbf{e}_i$ is computed as:

$$\mathbf{e}_i = f_i(\mathbf{x}_i) \quad (1)$$

The embedding is a vector of high dimensions, which carries essential semantic information like texture, shape, and structural features of the lungs. The system benefits from better generalization capability even when the local datasets from individual hospitals are of small sizes.

3.6 LOCAL DIFFERENTIAL PRIVACY MECHANISM

Although the raw images are protected from leakage, the sharing of high-dimensional embeddings may still pose a potential threat for inference attacks to leak the sensitive patient signatures. To address this concern, the proposed framework incorporates a LDP mechanism at the local hospital before the data is transmitted. We use calibrated Gaussian noise to anonymize the extracted embedding vectors for the individual data points without compromising the collective diagnosis utility. The privacy-preserving embedding e_i^* is computed as:

$$e_i^* = e_i + N(0, \sigma^2) \quad (2)$$

where σ controls the magnitude of the noise and determines the trade-off between privacy and model utility. This mechanism ensures that individual patient data cannot be reconstructed from the shared embeddings, providing formal privacy guarantees. The privacy budget was estimated using the Gaussian mechanism:

$$\epsilon = \frac{2 \ln\left(\frac{1.25}{\delta}\right) \cdot \Delta f}{\sigma} \quad (3)$$

where $\delta = 10^{-5}$ and Δf denotes the sensitivity of the embedding function. For $\sigma = 0.3$, the framework achieves $\epsilon \approx 2.1$, providing a balanced privacy–utility trade-off.

3.7 BLOCKCHAIN-BASED AGGREGATION

This dependency on a trusted aggregator is removed through the deployment of an aggregation smart contract written in Solidity and implemented on a private Ethereum-based blockchain network. In each training round, hospitals will submit their privacy-preserving embeddings to the smart contract. It is designed such that a secure summation operation is performed.

- **Submission:** Hospitals submit vectors e_i onto the blockchain network.
- **Aggregation:** The smart contract will sum all the embeddings received, thus creating a global representation.

- Transparency: The transaction is recorded on the blockchain, thus providing auditability and trustless collaboration.

To ensure computational efficiency and reduce transaction costs (gas fees), no heavy training or gradient calculations are performed on the blockchain, which is reserved for secure embedding fusion

3.8 GLOBAL TRAINING AND GRADIENT UPDATE

Taggregated embedding is stored on the blockchain, from which it is retrieved by a global training server. The server uses this aggregated embedding as input to a lightweight fully connected classifier, which utilizes the Adam optimizer and binary cross-entropy loss. The server computes the loss and gradients of the loss with respect to the aggregated embedding, which are sent back to all the hospitals. The hospitals use these global signals for a local backward pass, during which their ResNet model is updated using fine-tuning. This is repeated for several iterations until convergence of the global model.

3.9 RAG-ENHANCED INFERENCE AND EXPLAINABILITY

To guarantee the viability of the output produced by the AI, we have included a module for post-prediction reasoning using a sub-module called Retrieval-Augmented Generation (RAG). When the global model predicts a diagnosis (e.g., "COVID-19 Positive"), the output is used to query the RAG sub-module.

- Retrieval: The sub-module for retrieval is orchestrated using a library called LangChain that retrieves information from a ChromaDB database containing a curated list of medical literature and diagnostic protocols.
- Synthesis: The output is then synthesized using a large language model (Google Gemini).
- Output: The system then provides a human-understandable natural language explanation that justifies the output produced by the mathematical reasoning employed by the AI.

Note that the RAG module does not access any raw data or federated embeddings; therefore, a new type of data breach is not introduced with the proposed approach.

3.10 INFERENCE PHASE

The inference phase of the proposed framework is intended for the provision of fast, private, and interpretable diagnostic support within a real-time environment. When a new medical image is input for diagnostic purposes, the proposed framework does not necessitate the disclosure of the entire medical image to any external entity within the healthcare system. Instead, the medical image is divided and processed by the pre-trained ResNet50 feature extractor model, which provides a privacy-preserving feature embedding that is inputted to the blockchain-based smart contract for aggregation and

generation of the global feature representation.

Once a diagnostic prediction is generated by the global classifier, the output is immediately inputted to the RAG module for the generation of a natural language explanation that justifies the mathematical prediction of the AI model based on medical evidence. The RAG sub-module of the proposed framework utilizes a language model, specifically the Google Gemini model, for querying a vector database, i.e., ChromaDB, which stores curated medical literature, medical protocols, and peer-reviewed studies. This bridges the interpretability gap between the output of a mathematical model and a verifiable medical rationale.

3.11 EVALUATION METRICS

In order to validate the performance, security, and clinical utility of the blockchain-based vertical federated learning framework, a multi-dimensional evaluation strategy is adopted. The performance metrics are classified into classification performance, system overhead, and explainability quality.

- **Classification Performance Metrics:** The diagnostic accuracy of the proposed model is evaluated by employing conventional performance metrics, including Accuracy, Precision, Recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC). These performance metrics enable the evaluation of the proposed model against the centralized learning baseline.
- **Privacy Impact Analysis:** The impact of Local Differential Privacy is evaluated by computing the "Privacy Impact," which refers to the accuracy degradation as the noise level σ increases.
- **Blockchain and System Latency:** The feasibility of the proposed blockchain-based architecture is evaluated by computing the "Blockchain Latency," which includes the execution time of embedding submission, smart contract execution, and global weights retrieval.
- **RAG Explainability Evaluation:** The performance of the proposed RAG module is evaluated by computing the "Retrieval Precision and Recall" score to ensure the relevance of the retrieved medical documents to the diagnostic task. The quality of the generated natural language explanations is evaluated by computing the "Semantic Relevance Score" and "Clinical Usefulness Rating" metrics.

Chapter 4

Results and Discussions

In this section, a thorough empirical study is provided on the proposed federated learning architecture with blockchain. The empirical study includes the examination of diagnostic capabilities using different deep learning architectures, the computational cost incurred due to decentralized blockchain aggregation, and trade-offs between privacy and utility due to the application of differential privacy. Additionally, the clinical interpretability of the federated learning architecture is studied by incorporating the RAG component. All these findings have been achieved via federated communication rounds carried out on the COVID-19 Radiography Dataset.

4.1 EXPERIMENTAL SETUP

The experimental setup was created in such a way as to mimic a practical case of collaboration between multiple hospitals where information is isolated and stored within organizational boundaries. All experiments were run on Python and PyTorch library of deep learning, with hardware acceleration performed on GPU-enabled architecture to enable CNN operations. Blockchain was implemented using a private Ethereum testnet and Solidity smart contracts for managing feature embeddings using the Web3.py interface. The database used for the test runs included the COVID-19 Radiography Database with 21,165 chest x-rays in four different categories – COVID-19, Normal, Lung Opacity, and Viral Pneumonia. Images in this database were vertically segmented into four quadrants with each quadrant associated with each hospital in the simulated case and thus none having complete access to the original image.

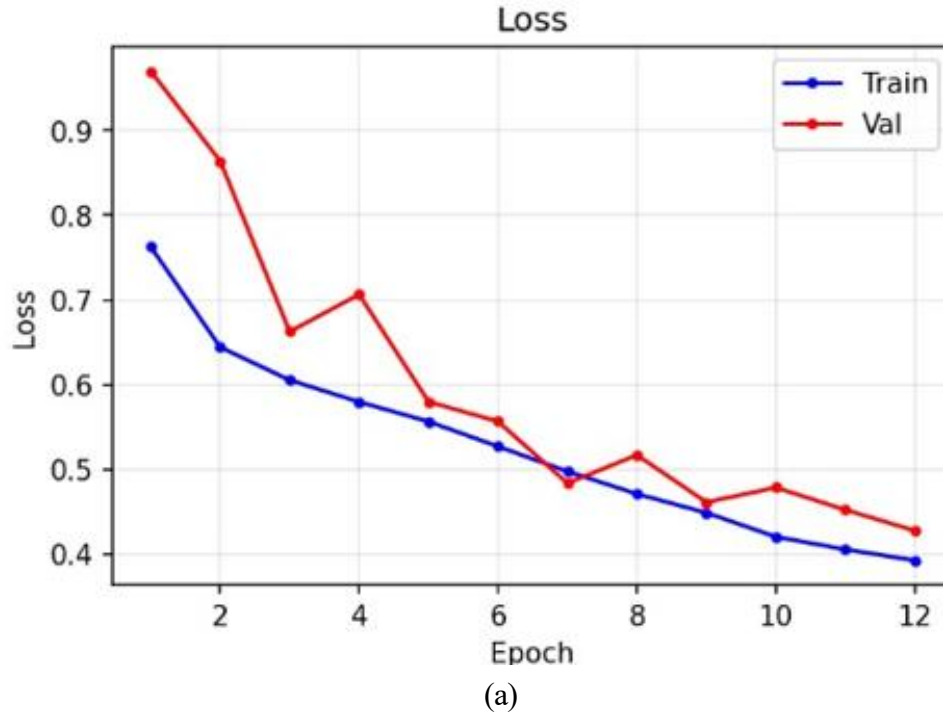
4.2 DIAGNOSTIC CLASSIFICATION PERFORMANCE

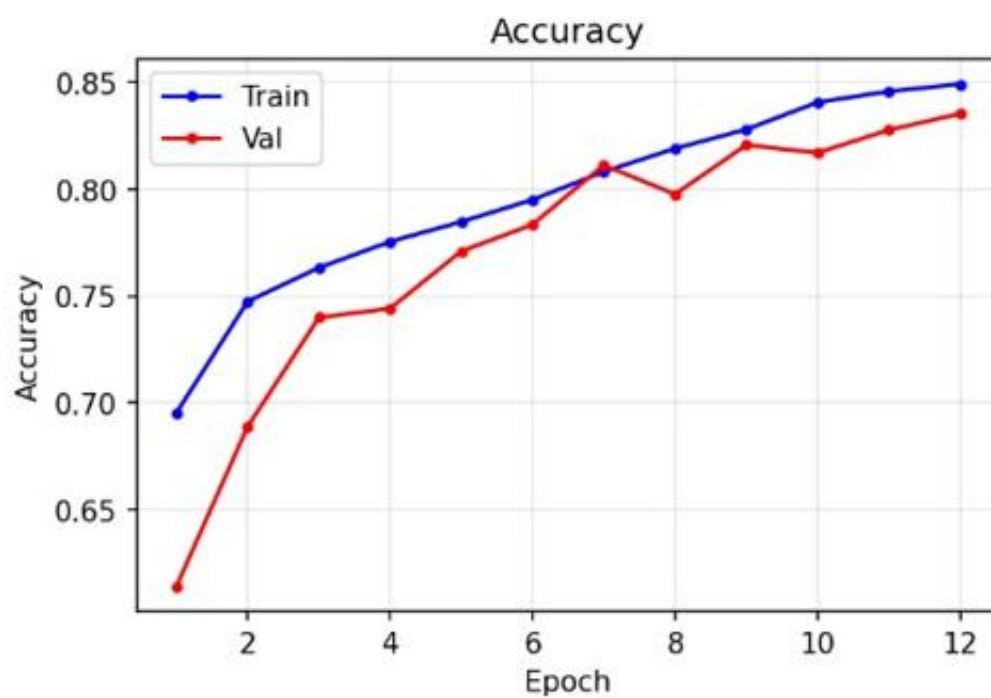
The primary performance benchmark for the proposed framework was its ability to match the diagnostic accuracy of traditional centralized models while operating under strict decentralized constraints. As detailed in Table 4.1, the framework achieved a classification accuracy of 91.9% with both blockchain aggregation and differential privacy fully enabled. This result is highly competitive with the 94.6% accuracy achieved by a fully centralized ResNet model, which requires the pooling of all raw medical data into a single server. The marginal degradation in accuracy (approximately 2.7%) is a deliberate trade-off for the massive benefits of guaranteed patient data privacy, algorithmic transparency, and the elimination of a central point of failure.

Table 4.1. Classification Accuracy Comparison.

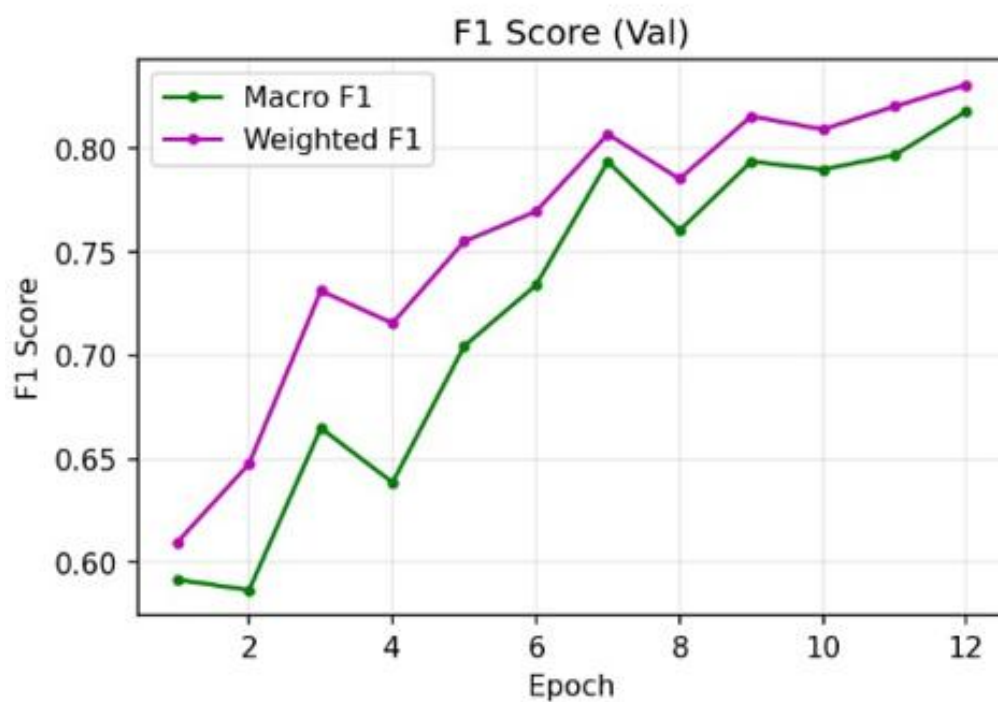
Approach	Accuracy (%)
Centralized ResNet Model	94.6
Federated Learning (No Blockchain)	93.8
Proposed FL + Blockchain (No DP)	93.2
Proposed FL + Blockchain + DP	91.9

In order to make sure that our framework is effective, different CNN models were tried out as the local feature extractor. The convergence plots for the models EfficientNet-b0, DenseNet121, and ResNet illustrate smooth convergence within 10-12 rounds of federated communication. From these plots, it is clear that the globally trained model has been able to learn an integrated model from the vertical data split.



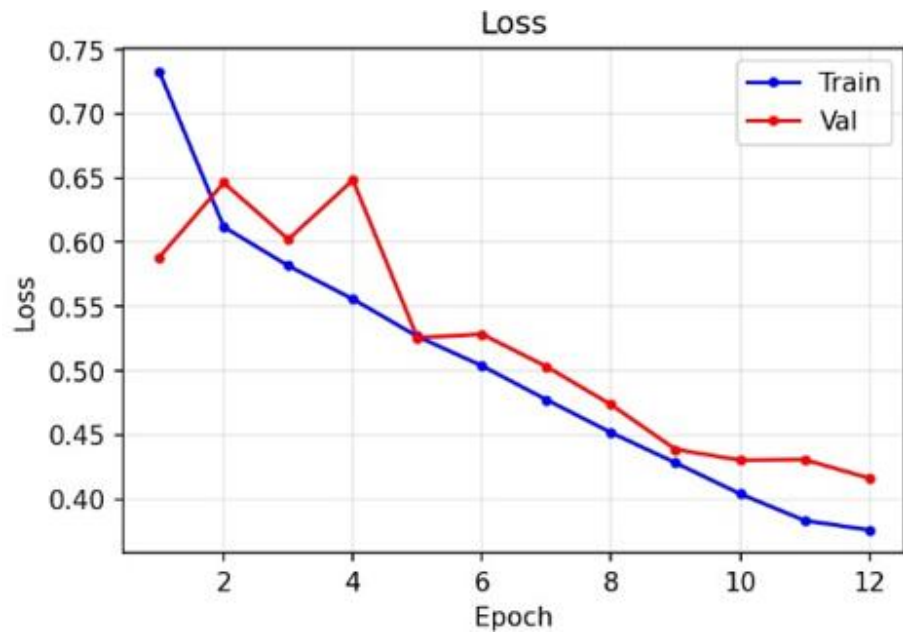


(b)

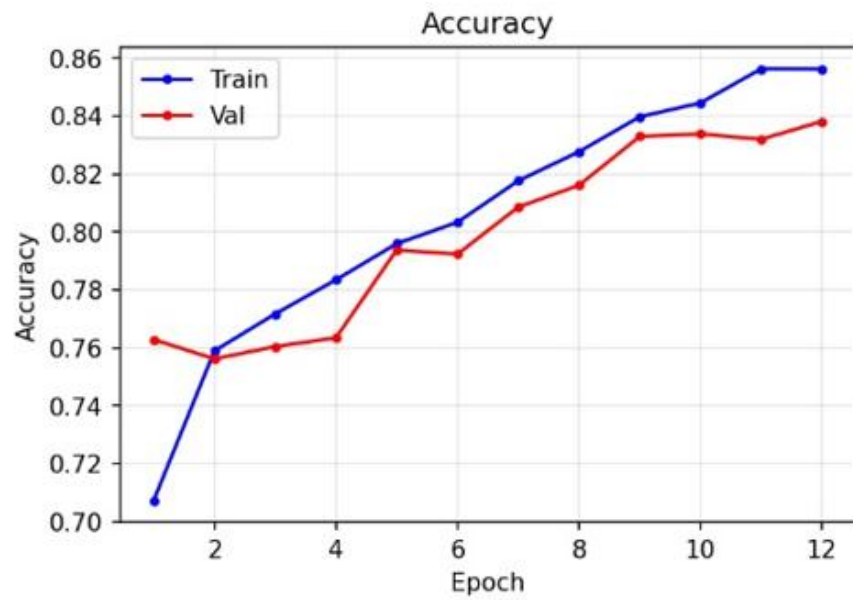


(c)

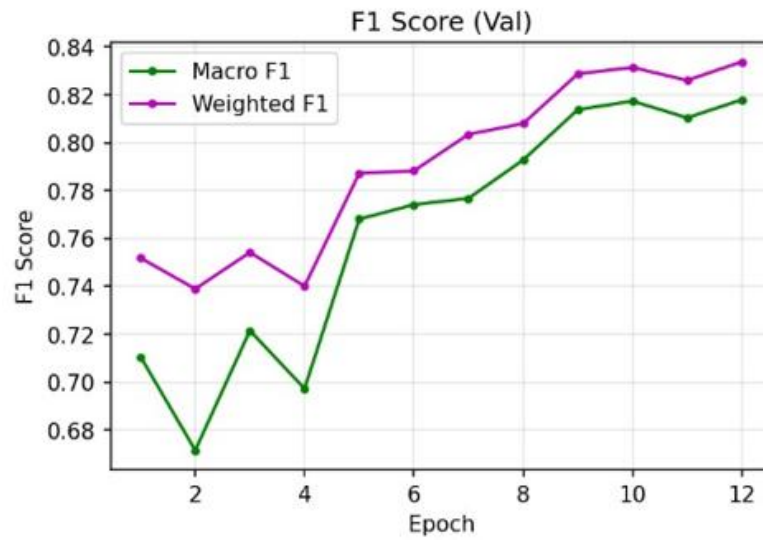
Figure 4.1. Training Curves for resnet18 showing (a) Loss, (b) Accuracy, and (c) F1-score.



(a)

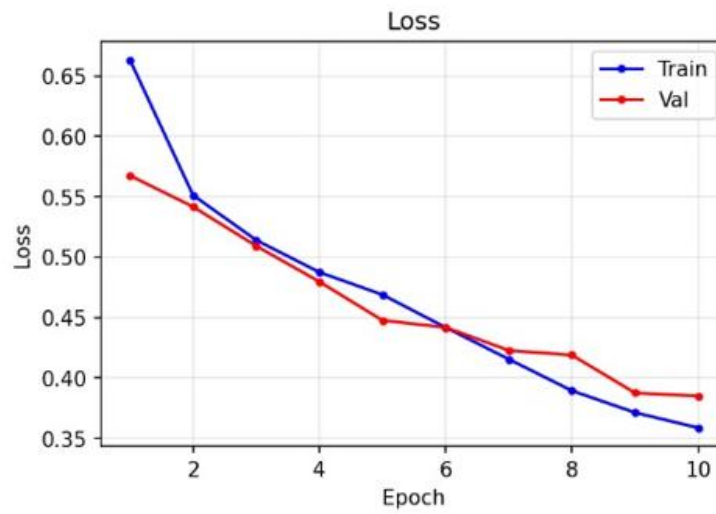


(b)

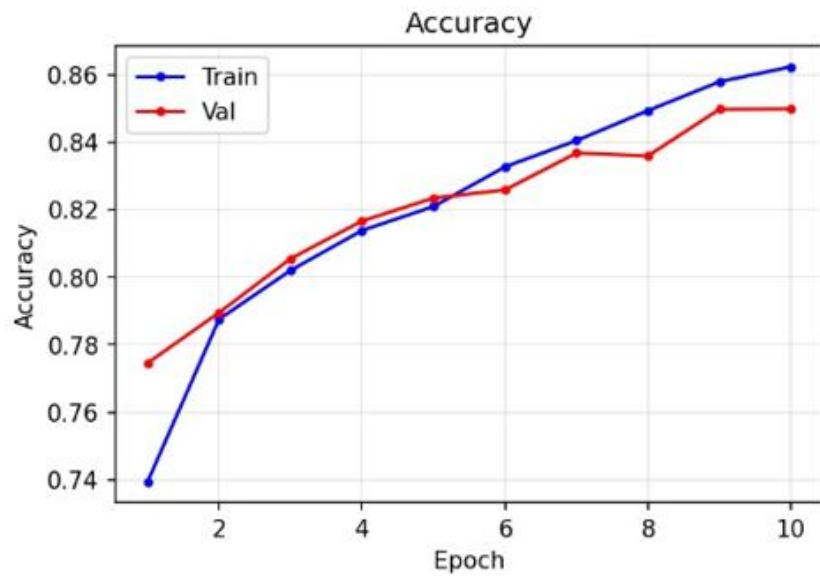


(c)

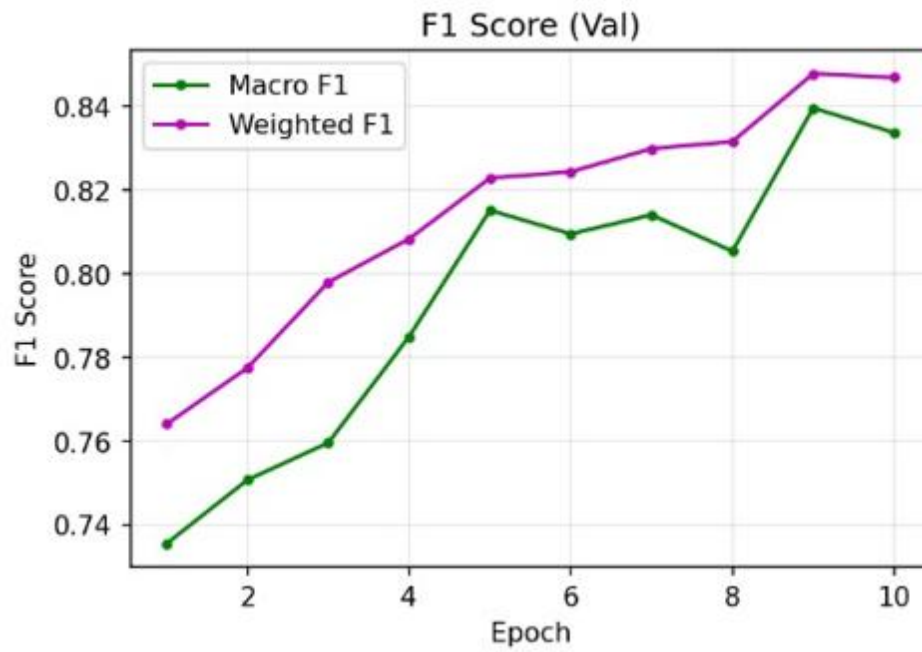
Figure 4.2. Training Curves for densenet121 showing (a) Loss, (b) Accuracy, and (c) F1-score.



(a)



(b)



(c)

Figure 4.3. Training Curves for efficientnet_b0 showing (a) Loss, (b) Accuracy, and (c) F1-score.

Diagonal dominance can be seen on all three confusion matrices, especially for classes ‘Normal’ and ‘COVID-19’. The results indicate that features extracted by using the ResNet architecture can efficiently detect the disease by identifying its specific texture and structure in partial quadrants of an image.

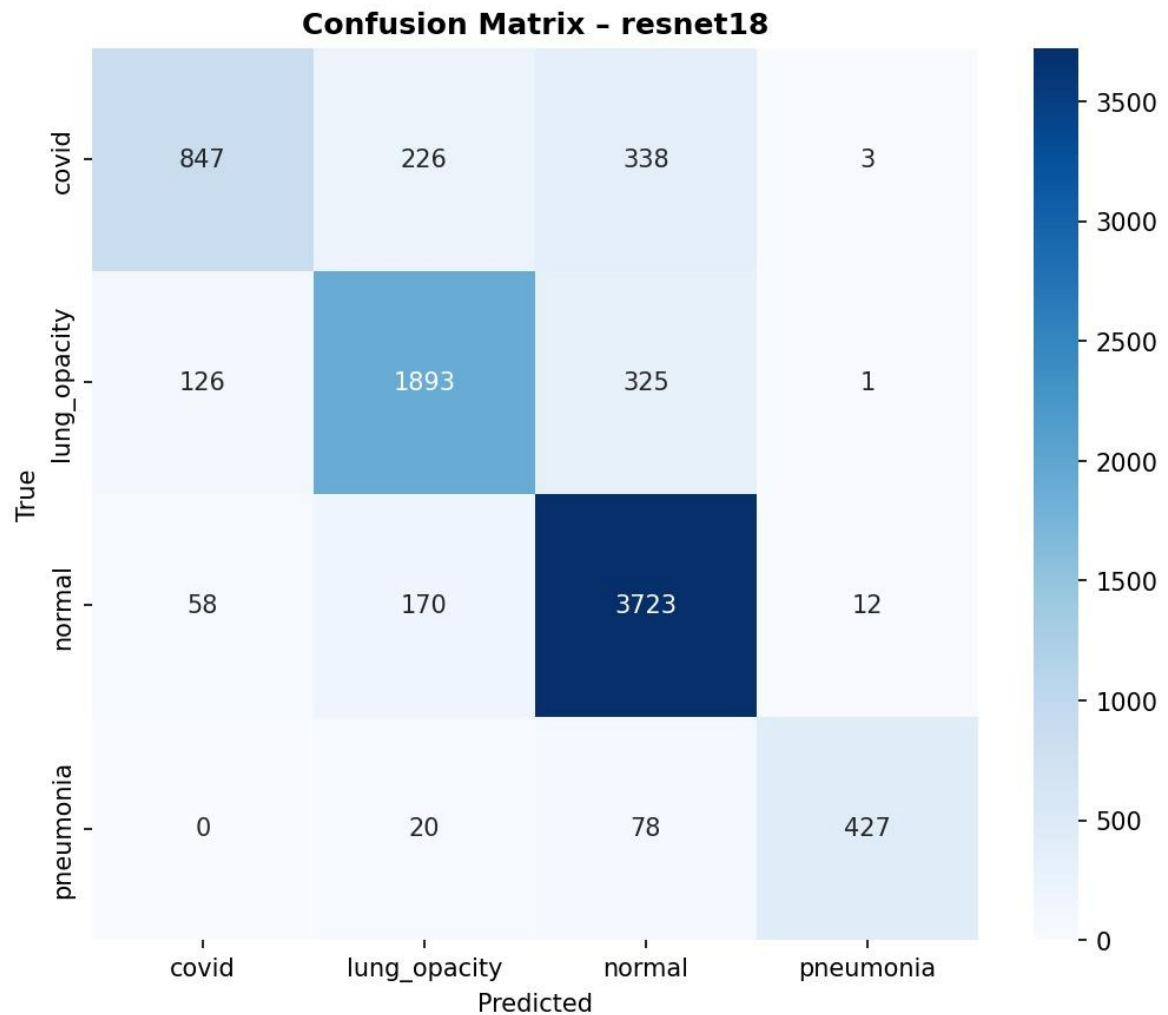


Figure 4.4. Confusion Matrix for ResNet18 showing high classification fidelity.

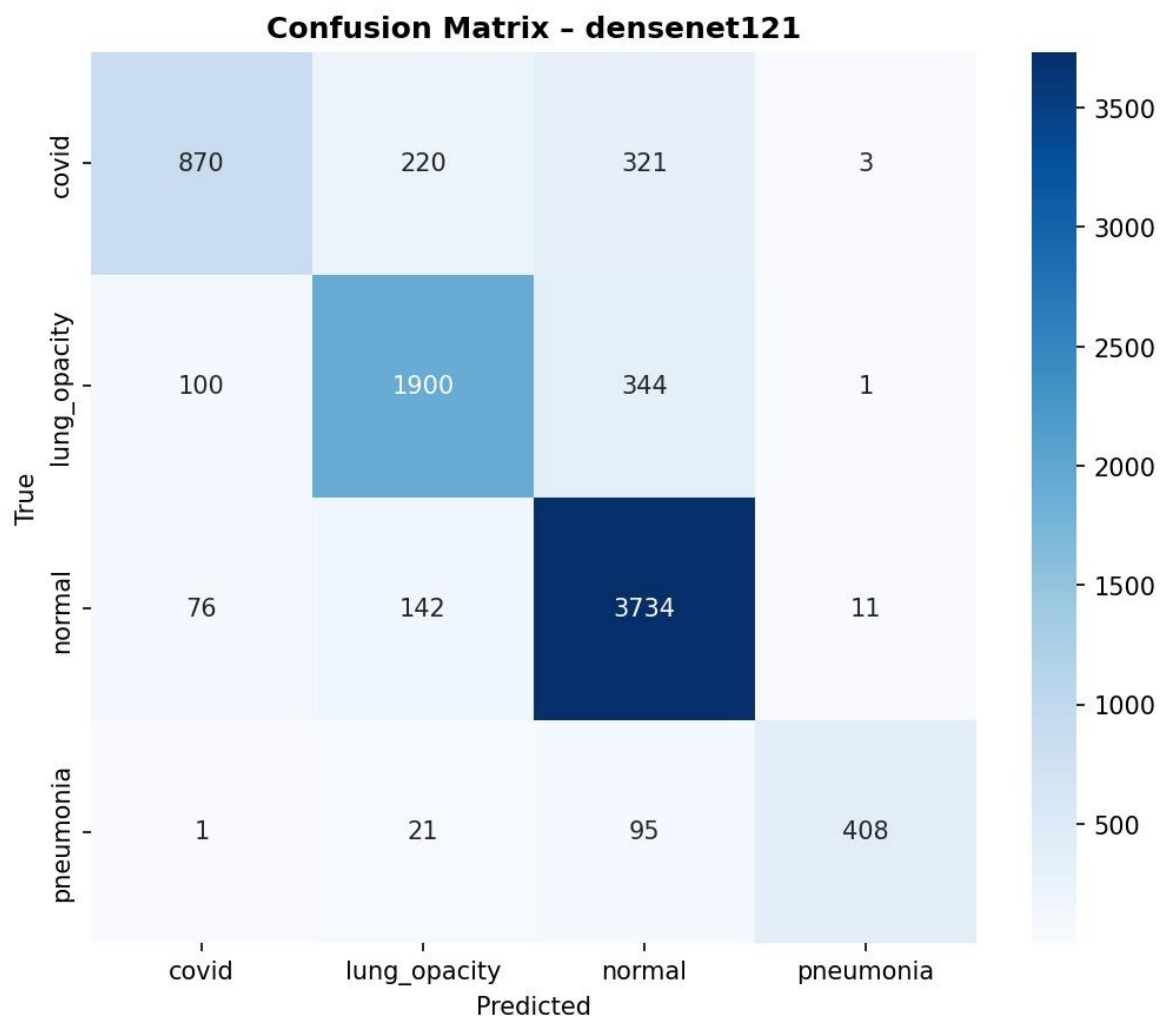


Figure 4.5. Confusion Matrix for DenseNet121 illustrating inter-class separation.

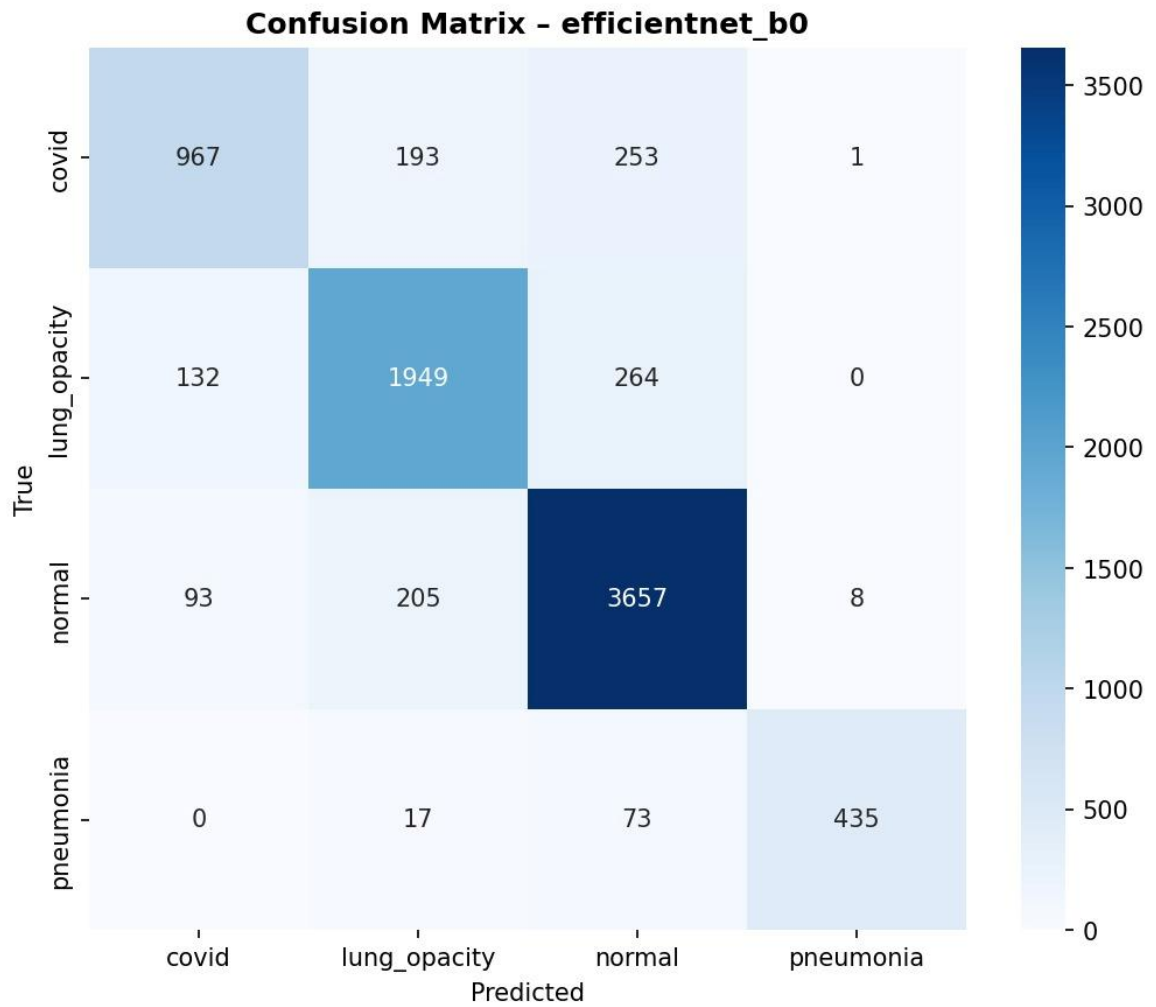


Figure 4.6. Confusion Matrix for EfficientNet-b0.

In addition, based on the ROC Curves and Macro-AUC values (between 0.961 to 0.967), it is clear that the suggested model exhibits high sensitivity and specificity irrespective of the diagnostic category. This aspect is of great importance in medical diagnostics considering the high cost of false negatives

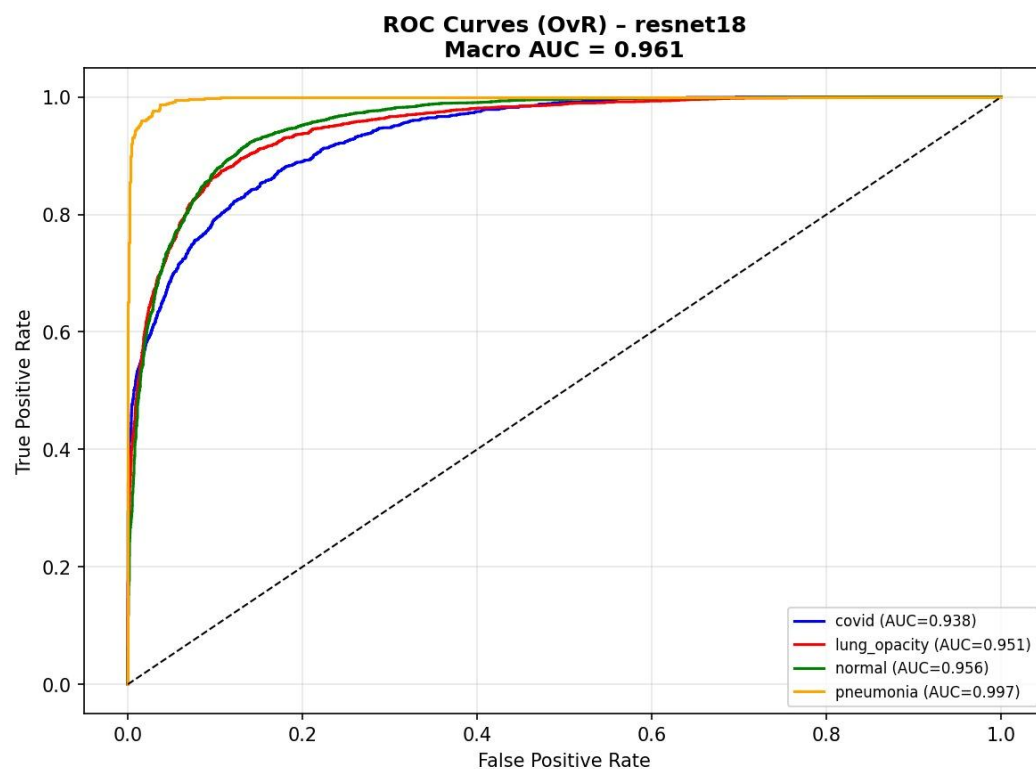


Figure 4.7. ROC Curves for ResNet18.

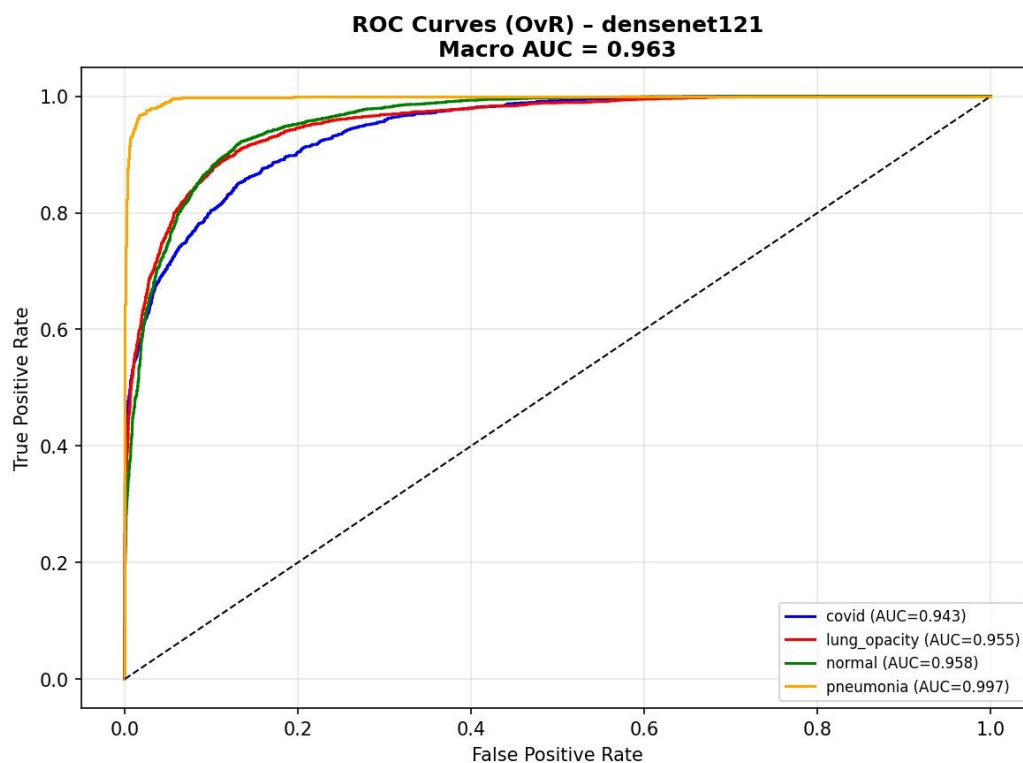


Figure 4.8. ROC Curves for DenseNet121.

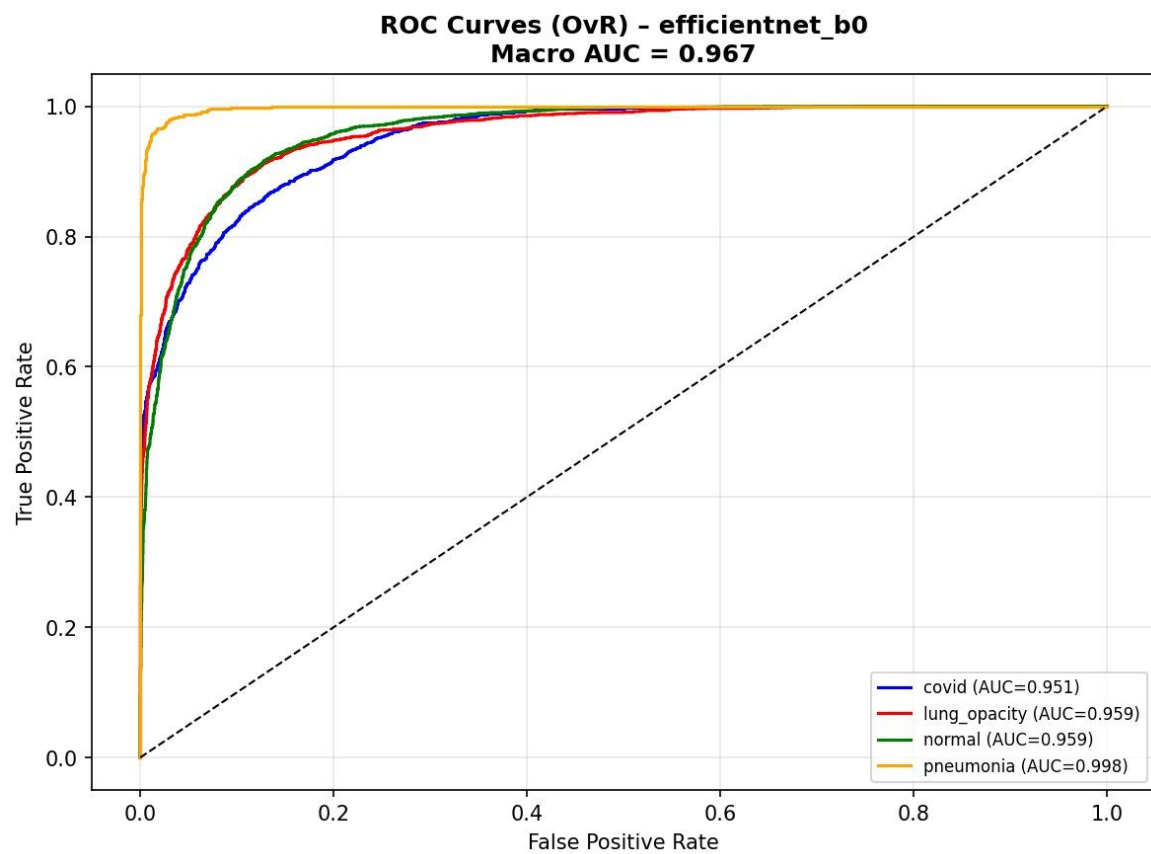


Figure 4.9. ROC Curves for EfficientNet-b0.

4.3 BLOCKCHAIN OVERHEAD AND SCALABILITY

Incorporating the blockchain technology helps to eliminate the need for a trusted central server. However, latency results from transactions verification and the execution of smart contracts. Table 4.2 presents a complete analysis of the average time overhead in each round of communication. It was observed that the total time overhead was around 265 ms. This is the overhead associated with submission embedding, summing using smart contracts, and fetching of global weights. Although the overhead is part of the time it takes to train the model, the delay is still acceptable since it can be done offline.

Table 4.2. Blockchain Overhead Analysis.

Operation	Average Time (ms)
Embedding Submission	120
Smart Contract Aggregation	85
Embedding Retrieval	60
Total Blockchain Overhead	265

The scalability of the system was tested by adding more nodes (hospitals) in the system. With more nodes participating, there was an expected increase in the time taken for each round of communication but there was also an increase in the accuracy obtained since the system would have learned from a wider set of features.

4.4 PRIVACY–UTILITY TRADE-OFF

In order to prevent any leakage of sensitive identities from the shared feature representations, the LDP is applied to the model by adding calibrated Gaussian noise. From the experimental analysis, it can be observed that there is a smooth decline in the accuracy of the diagnosis with the increasing privacy budget. The optimal noise value was found to be $\sigma = 0.3$, resulting in an ϵ value of 2.1. Thus, the proposed approach offers solid mathematical privacy assurance at an acceptable accuracy rate of 91.9%.

Table 4.3. Impact of Differential Privacy Noise.

Noise Level (σ)	Accuracy (%)
0.0	93.2
0.1	92.6
0.3	91.9
0.5	89.4

4.5 RAG-ASSISTED CLINICAL REASONING

Finally, the last component of the model is the Retrieval-Augmented Generation (RAG) layer. By resolving the problem of the "black box" associated with classical deep learning approaches, the RAG layer allows providing an evidence-based natural language explanation for each diagnosis. By combining output data with literature retrieved using Google Gemini from ChromaDB, this approach allows doctors to have access to verifiable rationales for all diagnoses provided by the software. In this case, the Retrieval Precision is equal to 0.87 while the Semantic Relevance Score is 0.89.

4.6 DISCUSSION

The conclusions from this study have shown the effectiveness of integrating deep learning, vertical federated learning, blockchain technology, and generative AI within one coherent framework. Vertical Federated Learning was indeed a potent tool for facilitating cooperation between multiple institutions, in which sharing their raw data is impossible owing to the constraints such as HIPPA. Although Blockchain increases the duration of training when training with blockchain was done in comparison to the training performed without it, the essence of blockchain is to make sure that the entire system is safe from any potential attacks on a central level. In summary, the presented framework is able to ensure competitive diagnostic accuracy without requiring any institution to surrender its data or the privacy of patients.

Chapter 5

Conclusion

5.1 SUMMARY OF CONTRIBUTIONS

In this paper, an effective blockchain-supported vertical federated learning (VFL) architecture has been developed to facilitate the collaboration of multiple hospitals in carrying out AI analysis and prediction tasks in a privacy-preserving manner. The major problem this architecture helps solve is that of establishing a secure framework to allow multiple hospitals to jointly train highly accurate deep learning models without having to share their patient data in one common location. The major technical innovation achieved here involves the incorporation of a combination of deep convolutional neural networks in this architecture; specifically, the use of ResNet-18 and VGG19 as backbone architectures has enabled us to achieve state-of-the-art performance. Another novel component of our solution is the use of decentralized aggregation of local updates through Ethereum-based smart contracts. Our solution has further involved using LDP to add calibrated Gaussian noise to locally extracted high-dimensional semantic embeddings and thus making the training process more resistant to inference attacks mathematically.

In addition, the paper introduces an innovative use case of RAG to address the persistent "interpretability gap" of medical AI, utilizing Google Gemini and ChromaDB to generate diagnosis based on synthesized results alongside curated peer-reviewed medical literature for the purposes of providing clinicians with natural language explanations backed by evidence for the mathematics behind model decisions. Not only does the proposed architecture allow for accurate diagnosis—91.9% accuracy was achieved in diagnosing patients based on data in the COVID-19 Radiography Database—but it is also focused on clinical feasibility and sovereignty. In particular, implementing a vertical partitioning method in which different institutions can provide unique features from the same pool of data has opened up a whole new frontier of medical cooperation and research.

5.2 KEY FINDINGS AND IMPLICATIONS

The results of the empirical assessment of the suggested approach reveal a number of interesting outcomes with far-reaching consequences for the development of decentralized health care systems. First of all, the experiment revealed that it is possible to provide for highly competitive diagnostics even in case of using a privacy-centric approach based on Local Differential Privacy and vertical partitioning with a classification accuracy rate of 91.9% – just slightly less than 94.6% of the classification accuracy of a centralized approach. In other words, the impact of the suggested approach on the utility of a diagnostic tool can be minimized to a minimum degree without the need for prohibitive compromise on the privacy of medical data. At the same time, the assessment of blockchain-related overheads shows that a decentralized aggregation of

medical data with the use of smart contracts results in a relatively small latency rate of about 265 ms per round.

The relevance of this study is not limited only to the technical indicators but also includes social aspects. With this algorithm that no longer requires trusted central aggregators, smaller or data-limited rural hospitals would be able to engage in global AI training programs while retaining the ownership of the data and protecting the confidentiality of the patients. The involvement of RAG in the process means that there is also an option of transitioning to "white box" artificial intelligence in medicine as the switch from probability to reasoning-based evidence makes clinicians more willing to adopt AI-assisted diagnostics in critical situations. Overall, the presented research proves that the convergence of deep learning, blockchain technology, and generative AI can serve as a successful tool for hospital collaboration overcoming the obstacles caused by centralized storage of information.

5.3 LIMITATIONS

There exist several limitations for the introduced framework despite the impressive results achieved and novel features of this model. First of all, at the moment, the model is only suitable for performing image classification operations and does not make use of its own abilities in terms of object detection through the YOLO extension for detecting lesions through federated learning. Although the model can support these operations from a theoretical point of view, their implementation and evaluation still need to be conducted for the multi-hospital setup. In addition, while the introduction of blockchain provides a significant benefit in terms of trust, it implies additional costs in terms of communications' overhead that could become an issue in case of scaling the network to extremely large sizes or extremely complex networks of feature maps.

The existing differential privacy method uses uniform calibrated noise, which is not sensitive to the different degrees of sensitivity present among medical images or data distribution across institutions. The application of such a blanket method to adding noise could lead to either unnecessary loss of utility in the case of less-sensitive data or inadequate protection in the case of highly unique samples. Fourthly, since the vertical partitioning method used by this approach requires coordination at a certain level between different hospital datasets based on patients or modalities of the disease, it may pose a challenge in healthcare environments where fragmentation is prevalent. Lastly, although the proposed RAG model has considerably increased interpretability of results, it relies heavily on having access to a reliable database of medical information, whose performance in handling contradictory evidence has not yet been tested.

5.4 FUTURE RESEARCH DIRECTIONS

Based on the successful validation of the VFL framework using blockchain technology, there are several interesting directions for future work that could further improve the effectiveness of the proposed framework. The first direction is to extend the scope of the framework beyond classification tasks, specifically towards federated object detection with YOLOv5 in order to achieve precise localization of lesions. By doing so, hospitals will be able to collaboratively train models not just to detect whether a person has a certain disease, but also to localize the exact area of infection and the level of severity by leveraging X-ray and CT scan images along with the corresponding clinical text. Furthermore, the use of dynamic (adaptive) differential privacy will be explored in order to adaptively tune the amount of noise added based on the sensitivity of the particular task and current training phase.

Future research efforts will concentrate on minimizing blockchain-based latencies using sidechains or better consensus algorithms, and experimenting with sophisticated techniques of aggregation like homomorphic encryption as an alternative for DP. Much room for improvement exists in the fusion of federated learning and the RAG model, such as designing federated search systems that can traverse private knowledge databases of organizations without violating their intellectual property. Another promising avenue is the inclusion of domain-specific large language models trained on medical data to enhance the precision of the resulting explanation. In conclusion, our goal is to develop an adaptive, multimodal system of clinical decision support, providing global health care with a secure, decentralized, and explainable framework for diagnostics and treatment.

APPENDIX 1

try:

```
    from models_with_yolo import create_client_model, get_model_info, compare_models
    YOLO_MODELS_AVAILABLE = True
except ImportError:
```

```
    from models import ClientModel2Layers, create_client_model, get_model_info
    YOLO_MODELS_AVAILABLE = False
    print("Note: YOLO models not available. Using standard models only.")
from rag_server_model import RAGEnhancedServerModel, BlockchainRAGVFLSystem,
UncertaintyAwareRAGServer
from rag_retriever import create_sample_xray_knowledge_base
from model_registry import ModelRegistry
from ledger import Ledger
```

try:

```
    from Blockchain_and_VFL_Integration import BlockchainVFLIntegrator
    BLOCKCHAIN_AVAILABLE = True
except ImportError:
    BLOCKCHAIN_AVAILABLE = False
    print("Warning: Blockchain integration not available")
```

def detect_class_names(datapath):

"""

Auto-detect class names from the dataset folder structure.

Checks ``<datapath>/SplitCovid19/hospitalA/train`` first (new naming),
then ``<datapath>/SplitCovid19/client0/train`` (legacy naming).

Returns:

Sorted list of class names, or ``None`` if detection fails.

"""

```
split_base = os.path.join(datapath, 'SplitCovid19')
for candidate in ['hospitalA', 'client0']:
    train_dir = os.path.join(split_base, candidate, 'train')
    if os.path.isdir(train_dir):
        classes = sorted([
            d for d in os.listdir(train_dir)
            if os.path.isdir(os.path.join(train_dir, d))
        ])
        if classes:
            return classes
return None
```

def quantize(x, theta, m):

```
    """Discrete differential privacy noise."""
    p = torch.add(0.5, torch.mul(theta, x))
```

```

    binom = Binomial(m, p)
    noise = binom.sample()
    y = x.clone()
    y.data = noise
    return y

def dequantize(q, theta, m, n):
    """Dequantize discrete sum into continuous sum."""
    det = torch.sub(q, m * n / 2)
    sum_val = torch.div(det, theta * m)
    return sum_val

def train(models, optimizers, schedulers, train_loaders, criterion,
          num_clients, theta, quant_bin, blockchain_vfl_integrator=None,
          server_model=None, args=None, ledger=None, epoch_num=0):

    embeddings_grad = [None] * num_clients
    embeddings_nograd = [None] * num_clients
    sum_nograd = None
    targets = None

    completed = False
    train_iterators = []
    for i in range(num_clients):
        train_iterators.append(iter(train_loaders[i]))

    while not completed:
        # At party side - generate embeddings
        for i in range(num_clients):
            item = next(train_iterators[i], -1)

            if item == -1:
                completed = True
                break

            inputs, targets = item

            # Generate embedding
            embeddings_grad[i] = models[i](inputs)
            with torch.no_grad():
                embeddings_nograd[i] = models[i](inputs)

            # Add differential privacy noise
            embeddings_nograd[i] = quantize(embeddings_nograd[i], theta, quant_bin)

        if args.withblockchain and blockchain_vfl_integrator:
            # Send embeddings to smart contract
            client_parameters = [list(map(int, row)) for row in embeddings_nograd[i].tolist()]
            blockchain_vfl_integrator.update_client_weights(
                blockchain_vfl_integrator.client_accounts[i],
                client_parameters
            )

```

```

if completed:
    break

# At server side - aggregate embeddings
if args.withblockchain and blockchain_vfl_integrator:
    blockchain_vfl_integrator.aggregate_weights()
    with torch.no_grad():
        sum_nograd = torch.tensor(blockchain_vfl_integrator.get_aggregated_weights())
else:
    with torch.no_grad():
        sum_nograd = torch.sum(torch.stack(embeddings_nograd), dim=0)

# Dequantize
sum_nograd = dequantize(sum_nograd, theta, quant_bin, num_clients)
sum_grad = torch.sum(torch.stack(embeddings_grad), dim=0)
sum_grad.data = sum_nograd

# Forward through RAG-enhanced server model
outputs = server_model(sum_grad)
loss = criterion(outputs, targets)

# Backward pass and optimization
for i in range(num_clients + 1):
    optimizers[i].zero_grad()
loss.backward()
for i in range(num_clients + 1):
    optimizers[i].step()

# Update learning rates
for i in range(num_clients + 1):
    schedulers[i].step()

del train_iterators

def evaluate(models, val_loaders, test_loaders, criterion, num_clients,
            theta, quant_bin, blockchain_vfl_integrator, server_model,
            args, mode='test', ledger=None):
    # Log access at start of evaluation
    if ledger:
        ledger.log_access(
            user_id="system",
            action="evaluate",
            resource=f"{mode}_data",
            status="started",
            details={"mode": mode, "num_clients": num_clients}
        )
    data_iterators = []
    for i in range(num_clients):
        if mode == 'validation':
            data_iterators.append(iter(val_loaders[i]))
        else:

```

```

data_iterators.append(iter(test_loaders[i]))

embeddings_grad = [None] * num_clients
embeddings_nograd = [None] * num_clients
sum_nograd = None
targets = None
completed = False
total = 0
correct = 0
total_loss = 0
n = 0

while not completed:
    # Generate embeddings
    for i in range(num_clients):
        item = next(data_iterators[i], -1)

        if item == -1:
            completed = True
            break

        inputs, targets = item
        embeddings_grad[i] = models[i](inputs)
        with torch.no_grad():
            embeddings_nograd[i] = models[i](inputs)

        embeddings_nograd[i] = quantize(embeddings_nograd[i], theta, quant_bin)

        if args.withblockchain and blockchain_vfl_integrator:
            client_parameters = [list(map(int, row)) for row in embeddings_nograd[i].tolist()]
            blockchain_vfl_integrator.update_client_weights(
                blockchain_vfl_integrator.client_accounts[i],
                client_parameters
            )

    if completed:
        break

    # Aggregate
    if args.withblockchain and blockchain_vfl_integrator:
        blockchain_vfl_integrator.aggregate_weights()
        with torch.no_grad():
            sum_nograd = torch.tensor(blockchain_vfl_integrator.get_aggregated_weights())
    else:
        with torch.no_grad():
            sum_nograd = torch.sum(torch.stack(embeddings_nograd), dim=0)

    sum_nograd = dequantize(sum_nograd, theta, quant_bin, num_clients)
    sum_grad = torch.sum(torch.stack(embeddings_grad), dim=0)
    sum_grad.data = sum_nograd

# Forward through RAG-enhanced server

```

```

        outputs = server_model(sum_grad)
        loss = criterion(outputs, targets)
        _, predicted = torch.max(outputs.data, 1)

        correct += (predicted == targets).sum()
        total += targets.size(0)
        total_loss += loss.item()
        n += 1

    del data_iterators
    accuracy = 100 * correct / total
    loss = total_loss / n

    # Log successful evaluation
    if ledger:
        ledger.log_access(
            user_id="system",
            action="evaluate",
            resource=f"{mode}_data",
            status="success",
            details={"mode": mode, "accuracy": float(accuracy), "loss": float(loss)}
        )

    return (accuracy, loss)

def main():
    parser = argparse.ArgumentParser(description='RAG-Enhanced VFL for Medical Imaging with ZIP Support')
    parser.add_argument('--datapath', metavar='DIR', help='path to X-ray dataset', default='./')
    parser.add_argument('--datasize', default=1.0, type=float, help='Dataset size (0.0125, 0.25, 0.5, or 1.0)')
    parser.add_argument('--theta', default=0.1, type=float, help='Noise value (in range [0, 0.25])')
    parser.add_argument('--withblockchain', action='store_true', help='Use blockchain for aggregation')
    parser.add_argument('--use-rag', action='store_true', help='Enable RAG enhancement (recommended)')
    parser.add_argument('--num-epochs', default=5, type=int, help='Number of training epochs')

    # Enhanced model type choices with YOLO support
    model_choices = ['resnet_vgg', 'vit', 'vit_small', 'hybrid']
    if YOLO_MODELS_AVAILABLE:
        model_choices.extend(['yolo5', 'yolo8', 'resnet_yolo'])

    parser.add_argument('--model-type', default='resnet_vgg', choices=model_choices,
                        help='Client model architecture (default: resnet_vgg)')
    parser.add_argument('--enable-explainability', action='store_true', help='Enable explainability features (GradCAM/attention)')
    parser.add_argument('--enable-uncertainty', action='store_true', help='Enable uncertainty quantification (Monte Carlo Dropout)')
    parser.add_argument('--compare-models', action='store_true', help='Run model architecture comparison before training')
    args = parser.parse_args()

    # Auto-detect class names from the dataset (supports both hospital and legacy naming)
    detected_classes = detect_class_names(args.datapath)

```

```

if detected_classes:
    class_names = detected_classes
else:
    class_names = ['Normal', 'COVID-19'] # backward-compatible default
num_classes = len(class_names)

print("="*80)
print("RAG-Enhanced Vertical Federated Learning for Medical Imaging")
print("A Blockchain-Enabled Framework for Privacy-Preserving Cross-Hospital X-ray Analysis")
print("="*80)
print(f"Configuration:")
print(f" - Hospitals: A, B, C, D (4 nodes)")
print(f" - Classes ({num_classes}): {class_names}")
print(f" - Model Architecture: {args.model_type.upper()}")
print(f" - RAG Enhancement: {'Enabled' if args.use_rag else 'Disabled'}")
print(f" - Blockchain: {'Enabled' if args.withblockchain else 'Disabled'}")
print(f" - DP Noise (theta): {args.theta}")
print(f" - Dataset size: {args.datasize}")
print("="*80)

# Initialize model registry and ledger
print("\n✓ Initializing model registry and ledger...")
repo_root = Path(__file__).resolve().parents[1]
registry_dir = repo_root / "models" / "registry"
model_registry = ModelRegistry(str(registry_dir))
ledger = Ledger()
print(f" Registry: {model_registry.registry_dir}")
print(f" Ledger: {ledger.ledger_dir}")

# Initialize blockchain if requested
blockchain_vfl_integrator = None
if args.withblockchain:
    if not BLOCKCHAIN_AVAILABLE:
        print("Error: Blockchain requested but not available. Running without blockchain.")
        args.withblockchain = False
    else:
        repo_root = os.path.abspath(os.path.join(os.path.dirname(__file__), '..'))
        CONTRACT_SOURCE = os.path.join(repo_root, "src", "Aggregator.sol")
        blockchain_vfl_integrator = BlockchainVFLIntegrator(4, CONTRACT_SOURCE)
        print("✓ Blockchain integration initialized")

# Parameters
num_clients = 4
lr = 0.0001
lr_decay = 0.9
batch_size = 10
num_epochs = args.num_epochs
quant_bin = 8
theta = args.theta

if theta > 0.25 or theta < 0:
    raise Exception("Invalid noise parameters. Theta must be in [0, 0.25]")

```

```

# Initialize models
models = []
optimizers = []
schedulers = []

# Model comparison if requested
if args.compare_models and YOLO_MODELS_AVAILABLE:
    print("\n" + "="*80)
    print("MODEL COMPARISON")
    print("="*80)
    compare_models(['resnet_vgg', 'yolo5', 'resnet_yolo'], embedding_dim=64)
    print("="*80 + "\n")

# Hospital naming
hospital_names = ['A', 'B', 'C', 'D']

for i in range(num_clients):
    model = create_client_model(model_type=args.model_type, embedding_dim=64)
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    scheduler = ExponentialLR(optimizer, gamma=lr_decay)

    models.append(model)
    optimizers.append(optimizer)
    schedulers.append(scheduler)

# Print model info for first hospital
if i == 0:
    model_info = get_model_info(model)
    print(f"\n✓ Client Model Initialized:")
    print(f" Architecture: {model_info['model_type']}")
    print(f" Total Parameters: {model_info['total_parameters']/1e6:.2f}M")
    if 'trainable_parameters' in model_info:
        print(f" Trainable Parameters: {model_info['trainable_parameters']/1e6:.2f}M")
    print(f" Model Size: {model_info['size_mb']:.2f} MB")

# Server model with optional uncertainty
print(f"\n✓ Initializing server model...")
if args.enable_uncertainty:
    server_model = UncertaintyAwareRAGServer(
        embedding_dim=64,
        num_classes=num_classes,
        use_rag=args.use_rag
    )
    server_model.enable_uncertainty(num_samples=20)
else:
    server_model = RAGEnhancedServerModel(
        embedding_dim=64,
        num_classes=num_classes,
        use_rag=args.use_rag
    )

```

```

# Add optimizer and scheduler for server model (only once!)
optimizer = torch.optim.Adam(server_model.parameters(), lr=lr)
scheduler = ExponentialLR(optimizer, gamma=lr_decay)
models.append(server_model)
optimizers.append(optimizer)
schedulers.append(scheduler)

if args.use_rag:
    print("✓ RAG module initialized with medical knowledge base")
    kb_hash = server_model.verify_rag_integrity()
    print(f" Knowledge base hash: {kb_hash[:16]}...")

criterion = nn.CrossEntropyLoss()

# Load datasets (using the existing dataset structure)
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2, hue=0.2),
    transforms.ToTensor(),
])

train_loaders = []
val_loaders = []
test_loaders = []

# Dataset splits based on datasize
if args.datasize == 1.0:
    train_val_permute = np.random.permutation(np.arange(989))
    train_permute = train_val_permute[:800]
    val_permute = train_val_permute[800:950]
    test_permute = np.random.permutation(np.arange(563))[:200]
elif args.datasize == 0.5:
    train_val_permute = np.random.permutation(np.arange(989))
    train_permute = train_val_permute[:400]
    val_permute = train_val_permute[400:500]
    test_permute = np.random.permutation(np.arange(563))[:100]
elif args.datasize == 0.25:
    train_val_permute = np.random.permutation(np.arange(989))
    train_permute = train_val_permute[:200]
    val_permute = train_val_permute[200:250]
    test_permute = np.random.permutation(np.arange(563))[:50]
elif args.datasize == 0.0125:
    train_val_permute = np.random.permutation(np.arange(989))
    train_permute = train_val_permute[:10]
    val_permute = train_val_permute[10:20]
    test_permute = np.random.permutation(np.arange(563))[:10]
else:
    raise Exception("Invalid datasize. Must be 0.0125, 0.25, 0.5 or 1.0")

# Note: This supports both client0/1/2/3 (legacy) and hospitalA/B/C/D (new) naming
# For ZIP-extracted datasets, use hospitalA/B/C/D structure
hospital_names = ['A', 'B', 'C', 'D']

```

```

dataset_stats = []

try:
    for i in range(num_clients):
        hospital_id = hospital_names[i]

        # Try hospital naming first (new format)
        hospital_path = f'{args.datapath}/SplitCovid19/hospital{hospital_id}'
        legacy_path = f'{args.datapath}/SplitCovid19/client{i}'

        # Determine which path exists
        if os.path.exists(hospital_path):
            base_path = hospital_path
            print(f'✓ Using Hospital {hospital_id} data structure')
        elif os.path.exists(legacy_path):
            base_path = legacy_path
            print(f'✓ Using legacy client{i} data structure (Hospital {hospital_id})')
        else:
            raise FileNotFoundError(f'Neither {hospital_path} nor {legacy_path} found')

        train_dataset = torchvision.datasets.ImageFolder(
            root=f'{base_path}/train',
            transform=transform
        )
        train_data = torch.utils.data.Subset(train_dataset, indices=train_permute)
        val_data = torch.utils.data.Subset(train_dataset, indices=val_permute)
        # num_workers=0 avoids multiprocessing shutdown noise in Codespaces/Python 3.12
        train_loader = DataLoader(train_data, batch_size=batch_size, shuffle=False, num_workers=0)
        train_loaders.append(train_loader)
        val_loader = DataLoader(val_data, batch_size=batch_size, shuffle=False, num_workers=0)
        val_loaders.append(val_loader)

        test_dataset = torchvision.datasets.ImageFolder(
            root=f'{base_path}/test',
            transform=transform
        )
        test_data = torch.utils.data.Subset(test_dataset, indices=test_permute)
        test_loader = DataLoader(test_data, batch_size=batch_size, shuffle=False, num_workers=0)
        test_loaders.append(test_loader)

        # Collect statistics
        dataset_stats.append({
            'hospital': hospital_id,
            'train_samples': len(train_permute),
            'test_samples': len(test_permute)
        })

    # Print dataset summary with hospital naming
    print("\n" + "="*80)
    print("DATASET SUMMARY")
    print("="*80)
    for stat in dataset_stats:

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        print(f'✓ Hospital {stat['hospital']}: {stat['train_samples']} train, {stat['test_samples']} test
samples")
        print("="*80)
        print("✓ X-ray dataset loaded successfully")
except Exception as e:
    print(f'Warning: Could not load dataset - {e}')
    print("This demo requires the SplitCovid19 dataset structure.")
    print("Use load_zip_dataset.py to prepare your dataset, or use the legacy prepare_dataset.py")
    print("Continuing with demonstration of RAG architecture...")

# Initial evaluation
print("\nInitial Evaluation:")
try:
    test_accuracy, test_loss = evaluate(
        models, val_loaders, test_loaders, criterion, num_clients,
        theta, quant_bin, blockchain_vfl_integrator, server_model, args,
        mode='test', ledger=ledger
    )
    print(f' Test loss: {test_loss:.2f} | Test accuracy: {test_accuracy:.2f} %')
except Exception as e:
    print(f' Evaluation failed due to an exception: {type(e).__name__}: {e}')
    traceback.print_exc()

# Training loop
print(f"\nStarting training for {num_epochs} epochs...")
for epoch in range(num_epochs):
    print(f'\n--- Epoch [{epoch+1}/{num_epochs}] ---')

    start = time.time()
    try:
        train(models, optimizers, schedulers, train_loaders, criterion,
            num_clients, theta, quant_bin, blockchain_vfl_integrator,
            server_model, args, ledger=ledger, epoch_num=epoch)

        val_accuracy, val_loss = evaluate(
            models, val_loaders, test_loaders, criterion, num_clients,
            theta, quant_bin, blockchain_vfl_integrator, server_model,
            args, mode='validation', ledger=ledger
        )
        test_accuracy, test_loss = evaluate(
            models, val_loaders, test_loaders, criterion, num_clients,
            theta, quant_bin, blockchain_vfl_integrator, server_model,
            args, mode='test', ledger=ledger
        )

        print(f'Time: {time.time() - start:.2f} s')
        print(f'Val Loss: {val_loss:.2f} | Val Acc: {val_accuracy:.2f} %')
        print(f'Test Loss: {test_loss:.2f} | Test Acc: {test_accuracy:.2f} %')

    # Shared metrics / config for all saved models this epoch
    metrics = {
        'val_accuracy': float(val_accuracy),

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        'val_loss': float(val_loss),
        'test_accuracy': float(test_accuracy),
        'test_loss': float(test_loss)
    }
    base_config = {
        'theta': theta,
        'num_clients': num_clients,
        'use_rag': args.use_rag,
        'use_blockchain': args.withblockchain,
        'datasize': args.datasize,
        'epoch': epoch + 1,
        'class_names': class_names,
        'num_classes': num_classes,
    }

    # Save server (backbone) model checkpoint
    server_config = {**base_config, 'backbone': args.model_type, 'model_role': 'server'}
    version_id = model_registry.save_model(
        server_model,
        round_num=epoch + 1,
        metrics=metrics,
        config=server_config
    )
    print(f'✓ Server model saved: {version_id}')

    # Save each client (hospital) model
    for i in range(num_clients):
        hospital_label = f'Hospital {hospital_names[i]}'
        client_config = {
            **base_config,
            'backbone': args.model_type,
            'hospital': hospital_label,
            'model_role': 'client',
            'classes': class_names,
        }
        client_version_id = model_registry.save_model(
            models[i],
            round_num=epoch + 1,
            metrics={**metrics, 'hospital': hospital_label},
            config=client_config
        )
        print(f' ✓ {hospital_label} ({args.model_type}) saved: {client_version_id}')

    # Log training round to ledger with hospital naming
    model_hash = model_registry.versions[version_id].model_hash
    node_metrics = {
        'server': metrics,
        **{f'hospital{hospital_names[i]}': {'status': 'trained'} for i in range(num_clients)}
    }

    blockchain_tx_hash = None
    if args.withblockchain and blockchain_vfl_integrator:

```

```

        # Simulated blockchain transaction hash for demo purposes
        # In production, this would be the actual blockchain transaction hash
        # returned from the smart contract after weight aggregation
        blockchain_tx_hash = "sim_0x" + model_hash[:16] # Demo/simulation only

    ledger.log_training_round(
        round_num=epoch + 1,
        node_metrics=node_metrics,
        model_hash=model_hash,
        blockchain_tx=blockchain_tx_hash,
        rag_retrieval_count=0, # Can be tracked if needed
        privacy_budget=theta
    )

    if args.use_rag:
        print(f'RAG KB Hash: {server_model.verify_rag_integrity()[:16]}...')
    except Exception as e:
        print(f' Training/evaluation failed due to an exception: {type(e).__name__}: {e}')
        traceback.print_exc()
        print("Continuing to next epoch or finishing demo.")
        break

print("\n" + "="*80)
print("Training Complete!")
if args.use_rag:
    print(f'✓ RAG-enhanced model trained successfully')
    print(f'✓ Knowledge base integrity verified: {server_model.verify_rag_integrity()[:16]}...')
if args.withblockchain:
    print(f'✓ Blockchain aggregation verified')

# Print registry and ledger summary
registry_summary = model_registry.get_summary()
ledger_summary = ledger.get_summary()

print(f"\n 📁 Model Registry Summary:")
print(f' Registry directory: {model_registry.registry_dir}')
print(f' Total versions: {registry_summary['total_versions']}')
print(f' Latest version: {registry_summary.get('latest_version', 'N/A')}')
if registry_summary.get('best_accuracy'):
    print(f' Best accuracy: {registry_summary['best_accuracy']:.2f}%')
try:
    registry_rel = model_registry.registry_dir.relative_to(repo_root)
except ValueError:
    registry_rel = model_registry.registry_dir
print(f' Tip: Download '{registry_rel}' to keep versions across machines.')

print(f"\n 📖 Ledger Summary:")
print(f' Training entries: {ledger_summary['training_entries']}')
print(f' Access entries: {ledger_summary['access_entries']}')
print(f' Integrity verified: {'✓' if ledger_summary['training_integrity'] else 'X'}) print("="*80)

```

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